

Report Title: Long-Duration Energy Storage Technologies for High Renewable Penetration

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Summary

This research focuses on the problem of the disparity between energy production from renewable sources and electricity consumption within the UK and assesses the significance of long duration energy storage (LDES) technologies in mitigating this issue. The impetus for this research is the growing use of intermittent renewable energy sources, such as wind and solar power, causing multi day and extended duration discrepancies in supply and demand. The analysis was conducted using a technoeconomic method that entailed employing illustrative monthly data to determine demand and supply values, with the latter being computed by subtracting demand from supply. Four types of LDES systems were reviewed in terms of their technical performance, costs and operational features.

A continuous surplus shortage of energy is observed all throughout the year, with the surplus shortage reaching a maximum value of about 12,000-13,000 GWh during winters and decreasing to an approximate surplus shortage of 4,000-6,000 GWh during summers. Among the various technologies discussed, the lowest cost per GWh of stored energy (£45,000-£50,000/GWh) is observed in the cases of CAES and PHES, no single technology provides long duration storage at high efficiency. therefore, a combination of long duration storage technologies is required.

Overall, therefore, it has been found that no individual energy storage technology can solve the problem of the deficit/surplus energy problem for long periods of time. Rather, multiple LDES solutions need to be implemented.

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List of Abbreviations

LDES - Long-Duration Energy Storage

PHES - Pumped Hydro Energy Storage

CAES - Compressed Air Energy Storage

CAPEX - Capital Expenditure

O&M - Operation and Maintenance

UK - United Kingdom

GWh - Gigawatt-hour

MWh - Megawatt-hour

CO₂ - Carbon Dioxide

LCOE - Levelized Cost of Energy

1 Introduction

1.1 Background and Motivation

The use of low carbon-based energy systems has led to the widespread use of renewable sources of energy such as wind, solar and hydropower. The use of these sources of energy is crucial for the mitigation of greenhouse gases in the environment, hence the need to focus on decarbonization targets. Unlike other sources of energy, this form of energy is weather-dependent, hence bringing about several challenges that affect the maintenance of stability in the system.

In the United Kingdom, the use of renewable sources of energy for the generation of electricity in 2024 was over 50%. In 2024, wind was the major form of renewable energy.

1.2 Problem Statement and Knowledge Gap

One of the main problems associated with high renewable energy systems is the ongoing mismatch between renewable energy generation and electricity demand. For instance, in the UK, electricity demand increases during winter due to increased demand for heating and solar energy generation is zero, while wind energy generation may be variable over long periods. Although various solutions, such as short-duration energy storage, exist, they are not applicable to solving long-term energy imbalances, and hence fossil fuel-based backup and imports are still necessary, hindering the path to complete decarbonization.

Although various studies have been conducted on renewable energy and storage technologies, there is a lack of knowledge on the techno-economic evaluation of long-duration energy storage technologies for solving large and long-term energy imbalances in renewable-based energy systems.

1.3 Aim and Objectives.

To evaluate and compare different long-duration energy storage (LDES) technologies in terms of their environmental, financial and technical suitability for electrical systems with high renewable energy penetration.

The objective of this study is to determine the efficacy of long-duration energy storage technologies for the provision of power systems that are supplied with high levels of renewable energy sources.

The objectives are:

- To examine the level of supply/demand mismatch through monthly UK power supply/demand data
- To determine the gap in renewable power supply (in GWh/month) for different seasons
- To examine the efficacy of LDES technologies based on factors such as efficiency, cost and storage durations
- To determine the best technology for application, based on the efficacy results

The objectives are measurable and can be directly addressed in the results and discussion section.

1.4 Scope of the Study

The research has used the United Kingdom electricity system as a case study, considering its high levels of renewable energy penetration and its ambitious decarbonization targets. The research has a techno-economic and qualitative approach and uses secondary data from energy reports and literature.

The research analysis involves modelling electricity demand and renewable generation profiles, determining the supply-demand gap and assessing a few LDES technologies such as pumped hydro, compressed air and flow batteries

1.5 Renewable Generation Characteristics and Grid Integration Challenges

The global shift towards low carbon energy solutions has result in rapid development in production of renewable electricity. Technologies like solar, wind, hydropower and marine energy play a vital role in low carbon energy solutions due to their low greenhouse gas emissions and sustainability. Unlike conventional thermal power plants, where electricity generation is stable and dispatchable. most of the renewable energy technologies used for electricity generation suffer from variability and dependency on external weather conditions. These factors create considerable technical and operational difficulties in power system stability and reliability(D J Ward et al., 2012; International Energy Agency, 2013).

Renewable energy technologies like solar and wind energy face issues such as intermittency, nonsynchronous operation, spatial dependency and low controllability. Intermittency is one of the major issues in renewable energy technologies like solar and wind energy. These technologies face fluctuations in power generation depending on meteorological conditions. These fluctuations may occur over various timescales, ranging from minute scale variability due to cloud movement and wind turbulence to seasonal variability in solar and wind patterns. As the renewable penetration increases, these characteristics reduce system inertia, complicate frequency regulation and increase the need for balancing mechanisms(National Grid ESO, 2022).

The challenges faced by grid integration are largely because of the need to maintain equilibrium between supply and demand at real time. Electricity supply is required to be always maintained at a constant frequency and failure to do so may cause damage to the infrastructure and lead to power outages. In addition, the peak demand for electricity is usually during the evening and solar power is either negligible or absent during these times. This has therefore led to the need for flexible supply, demand response, interconnection and storage. Based on (Department for Energy Security and Net Zero, 2020) Long Duration Energy Storage (LDES) is therefore significant for addressing long periods of low renewable output, such as low wind for several consecutive days or low solar output for several months.

1.6 UK-Focused Renewable Energy Generation Scenario

1.6.1 Renewable Energy Resources in the United Kingdom

The United Kingdom has experienced a considerable change in its electricity generation pattern over the last two decades. According to the report(Department for Energy Security and Net Zero, 2024) There has been considerable growth in renewable energy sources particularly wind and solar photovoltaic systems. In the last few years, renewable energy sources have contributed 40 to 50 percent of the total electricity generated annually in the United Kingdom. The share of wind power is the largest among the renewable energy sources. Under favourable conditions, renewable energy sources sometimes generate more electricity than the national demand.

Despite the considerable growth in renewable energy sources, there are considerable short term and seasonal changes in the renewable energy output. The renewable energy output in the United Kingdom is mainly contributed by wind and solar energy, which are both weather-dependent(Met Office, 2022; International Energy Agency (IEA), 2023a). While there is considerable growth in offshore wind energy, its output is very weather dependent. Solar generation, although increasing continuously, remains highly dependent upon latitude and seasonal changes in irradiance and thus experiences considerable reductions in the winter months when electricity demand tends to be at its highest. The characteristics of variability in solar energy can be observed, where the daily generation profile for a representative day has been provided. Solar energy has been observed to have a predictable diurnal profile that increases in the mornings, peaks in the middle of the day, and reduces to zero at night. This has resulted in a structural mismatch between the supply and demand of electricity because the peak electricity demand in the UK tends to occur in the evenings when the sun has gone down(National Grid ESO, 2025).

The variability of wind generation is observed on various timescales. The wind speed can vary considerably between consecutive days(Jiang *et al.*, 2023). Long periods of low wind energy generation, also referred to as “wind droughts,” can also occur. During such times, renewable energy supply is considerably less than anticipated levels. The result of solar intermittency and wind variability is a challenge that increases when more renewable energy is added to the supply(International Energy Agency (IEA), 2023a). The diversity of various types of renewable energy can somewhat compensate for variability. However, concurrent low output is a significant issue. Excessive levels of renewable energy supply led to a corresponding increase in supply-demand imbalance probabilities.

1.6.2 Current UK Renewable Electricity Mix

As per data available on the website of various government agencies and industries. renewable energy sources contributed more than half of the United Kingdom electricity supply for the first time in 2024(Carbon Brief, 2025; Department for Energy Security and Net Zero, 2025). The data available on various websites indicate that THE renewable energy sources contributed 50.4% to the United Kingdom's electricity supply in 2024, while it was 46.5% in 2023.

The wind energy sector is seen to contribute a significant amount to the renewable energy supply. In 2024, wind energy contributed around 29.2-30% to the United Kingdom's electricity supply. This is seen to be the largest contributor to renewable energy supply. Of this, offshore wind contributed 17%, while onshore wind contributed 12-13%. The solar energy sector also contributes to renewable energy supply. The solar photovoltaic systems contributed around 5-6% to the United Kingdom's electricity supply. The hydro energy sector also contributed to renewable energy supply. The hydro energy sector contributed around 1.5-2% to the United Kingdom's electricity supply. The bio energy sector also contributed to renewable energy supply. The biomass energy contributed around 5-8% to the United Kingdom's electricity supply.

To summarise the 2024 UK renewable electricity generation share:

Renewable Source	Approx. Share (%) of UK Electricity Generation
Wind	29.2–30.0
Solar PV	5.0–6.0
Biomass/Bioenergy	5.0–8.0
Hydropower	1.5–2.0
Total Renewables	50.40%

Table 1.1: 2024 UK renewable electricity generation share

This data highlights that the UK’s electricity system has become **predominantly renewable-led**, with wind power alone often meeting around a **third of annual electricity demand**.

1.7 UK Renewable Energy Resources (Ordered by Contribution)

1.7.1 Wind Energy (Largest Contributor – 29–30%)

Operating Principles and System Characteristics

Wind power generation harnessing of the kinetic energy of the air using wind turbines. As the air blows over the turbine blades, the aerodynamic action causes the turbine to rotate, thus generating power. The power generated from the air is greatly influenced by the speed of the air. The power generated is proportional to the cube of the velocity of the air. According to the report (Joselin Herbert *et al.*, 2007) indicates that the power generated is greatly influenced by the speed of the air. The power generated may vary even with a slight change in the velocity of the air. Modern wind power generation, both onshore and offshore, has been established to work well over a wide range of wind speeds. However, the power generated is still weather-dependent. Offshore wind power generation has a high-capacity factor compared to onshore generation because of the consistent nature of the wind.

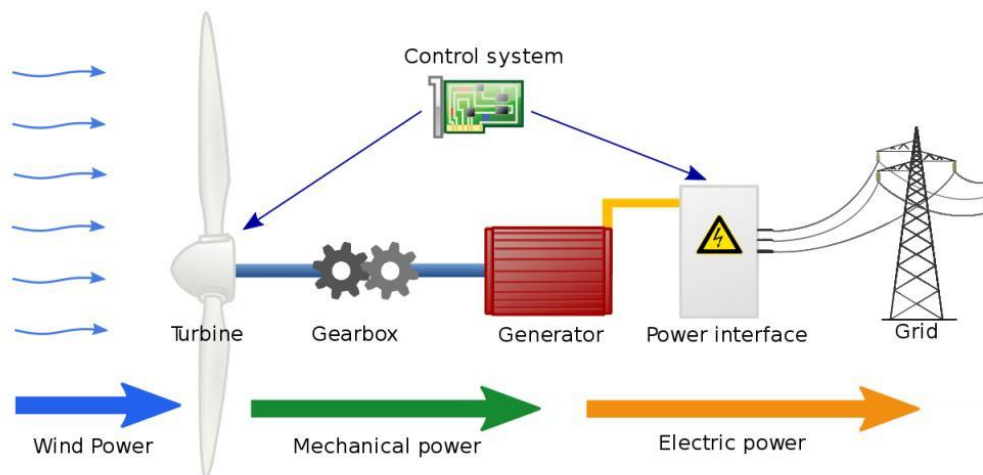


Figure 1.1: Schematic diagram of wind energy. source:(Burton, 2011)

Variability and Grid Implications

Yet, the wind energy generation is, by its very nature, variable over short and long-time scales. The wind speed varies hourly due to turbulence, and it varies seasonally according to the climatic conditions. Long stretches of low wind speed, lasting several days, can have a major impact on the energy generation capability over large areas. As the contribution of wind energy to the grid increases, the low energy generation capability over long stretches will have to be compensated by other forms of generation, thereby posing a challenge to the stability of the grid. Though wind energy plays a major role in the de-carbonization of the grid, the variability of the wind energy generation causes operational problems to the grid stability(Christopher *et al.*, 2015). Wind energy is presently the leading source of renewable energy used in electricity generation in the UK. In the year 2024(Department for Energy Security and Net Zero, 2025), wind energy contributed 29% to 30% to the

total electrical energy generated in the UK. It is also noted that wind energy is the most important source of renewable energy and, at times, the most important source of energy generation.

The UK has a high potential to generate energy through wind energy using offshore wind due to the high wind speeds in the North Sea, which enables energy to be generated at a high-capacity factor, i.e., between 40% to 55%. As wind energy is a major source of energy, contributing almost one-third to the total energy generated, it has a major impact on the stability of the grid due to its variability, which is highly dependent on meteorological conditions. Thus, wind energy is a major contributor to the need for long-duration storage. The figure indicates the monthly variation in electricity generated by wind in the United Kingdom. Electricity generated by wind power has a seasonal variation with higher electricity generated in the winter months and less in the summer months.

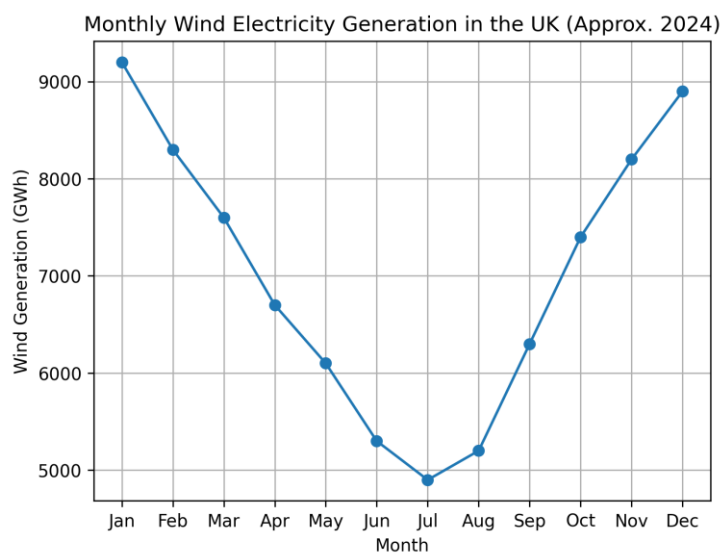


Figure 1.2: Monthly wind electricity generation in the UK

In the winter months, especially in December and January electricity generated by wind power is greater than 8,500-9,000 GWh due to increased Atlantic weather systems, resulting in higher average wind speeds in the United Kingdom. On the contrary, in the summer months, especially in June and July, the average wind speeds are low, resulting in less electricity being generated, especially less than 5,000-5,500 GWh.

This indicates that electricity generated by wind power is intermittent in nature, despite it being the highest source of electricity in the United Kingdom, **accounting for 29-30% of electricity generated in the United Kingdom**. Therefore, this variability indicates the need for long-duration energy storage (LDES) technologies. These technologies can store surplus electricity generated during high winds in winter and supply the same during times of lower wind energy generation. This will help to stabilize the grid and supply reliable electricity in a high renewable energy scenario.

1.7.2 Biomass and Bioenergy (5–8%)

Biomass is the second largest renewable energy source to contribute to the UK's electricity supply. On average, it supplies between **5% to 8% of the UK's overall electricity**(Department for Energy Security and Net Zero, 2025). Unlike wind farms and solar parks, the amount of electricity produced by biomass is dispatchable, it can be controlled according to the required amount.

A power station using biomass converts waste into steam to produce electricity. there are concerns surrounding the sustainability of the use of biomass. the contribution of biomass is to add reliability to the overall system by providing dispatchable renewable energy to offset the intermittency of wind and solar power(Rowe, Street and Taylor, 2009).

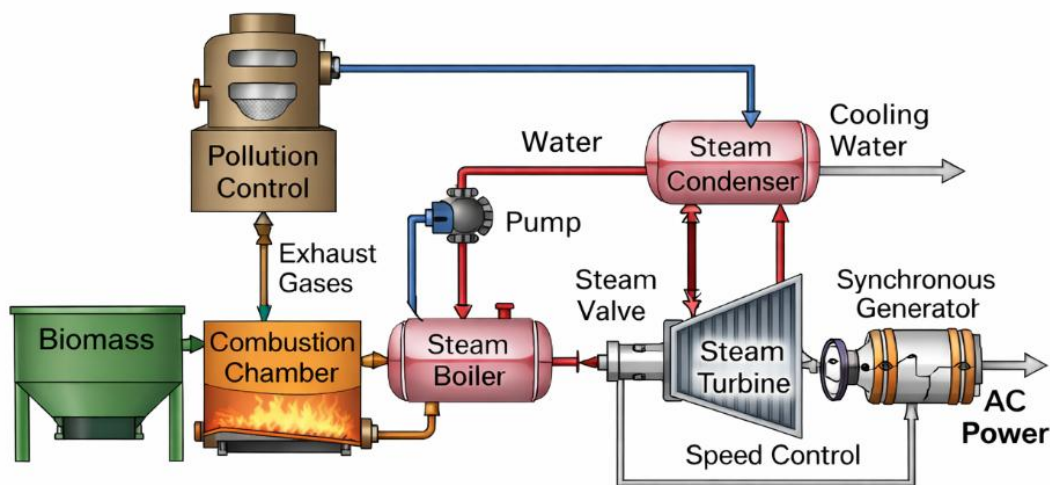


Figure 1.3: Schematic diagram of biomass power generation system. Source:(Demirbas and Ayhan, 2009)

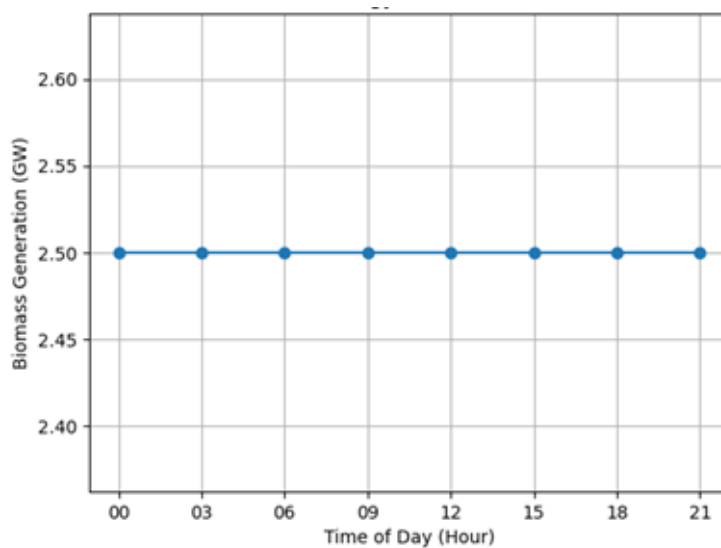


Figure 1.4: Monthly Biomass electricity generation in the UK

The biomass generation profile demonstrates a constant and consistent electricity output of about 2.5 GW over the course of the entire day. Biomass energy being dispatchable, can control the electricity output depending on the need.

The operation of a biomass power plant is almost like a traditional power plant, with the difference being the fact that it burns wood pellets or agricultural waste to produce electricity. Biomass, being dispatchable, can produce constant electricity output, making it a vital part of a renewable energy system, which can produce a counteractive effect to wind or solar power.

1.7.3 Solar energy (5–6%)

Solar Photovoltaic (PV)

Solar photovoltaic systems convert sunlight directly into electricity through the photovoltaic effect. When solar radiation strikes semiconductor materials (typically silicon), electrons are excited and generate direct current (DC) electricity, which is then converted to alternating current (AC) for grid use (International Energy Agency (IEA), 2023b).

Key Characteristics:

- **Diurnal Variation:** Solar generation occurs only during daylight hours.
- **Weather Dependency:** Cloud cover, atmospheric conditions, and shading significantly reduce output.
- **Seasonal Variation:** Solar irradiance is higher in summer and lower in winter in temperate climates.
- **Predictability:** Short-term forecasting is possible, but still uncertain.

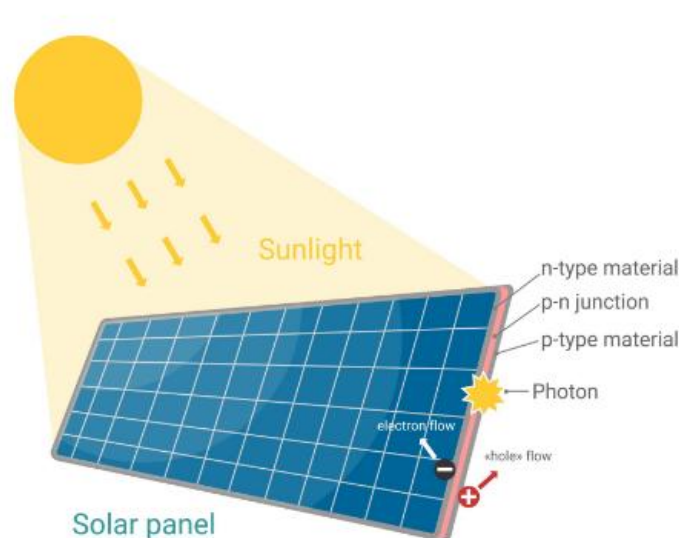


Figure 1.5: Schematic diagram of photovoltaic (solar panel)

The power output of a solar PV system can be approximated by:

$$P = \eta \cdot A \cdot G$$

P = Power output (W)

η = Panel efficiency

A = Panel area (m^2)

G = Solar irradiance (W/m^2)

Although solar energy is abundant and environmentally clean, its inability to generate electricity at night creates a fundamental mismatch between production and demand. In many countries, electricity demand peaks during the evening, when solar output is zero.

Concentrated Solar Power (CSP)

Concentrated Solar Power systems use mirrors to focus sunlight onto a receiver, producing heat that generates steam to drive turbines. Unlike PV systems, CSP plants can integrate thermal energy storage, enabling electricity generation beyond daylight hours. However, CSP deployment is geographically limited to high-irradiance regions and requires large land areas.

Solar PV accounts for approximately 5–6% of UK electricity generation (Department for Energy Security and Net Zero, 2025). While its annual contribution is lower than wind, solar output can reach very high instantaneous levels during sunny summer days. Solar energy follows a predictable daily pattern, with output peaking at midday and falling to zero at night. Seasonal variation is significant in the UK due to latitude, with winter generation substantially lower than summer output. Although solar contributes a smaller annual share compared to wind, its daily variability creates sharp ramping challenges for the grid, particularly during evening demand peaks.

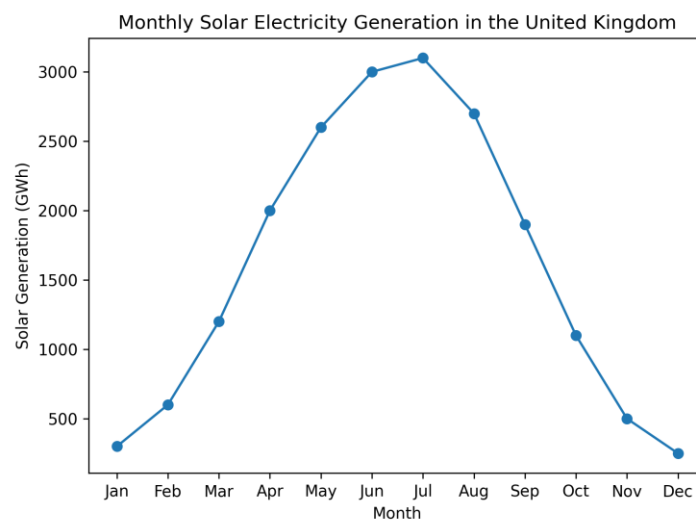


Figure 1.6: Monthly solar electricity generation in the UK

The solar energy generation starts to rise gradually in February and March as the days become longer in the spring season. The electricity production rises quickly in April and May as the solar irradiance levels are stronger and the days are longer. The solar electricity production peaks in June and July as solar electricity production rises to levels greater than 3000 GWh per month. This is because these months have the longest period of sunlight in the United Kingdom.

After the summer peak, the solar electricity production gradually falls in August and September as the days become shorter in the summer. This trend of solar electricity production falls in the autumn months, such as October and November, as the solar irradiance falls significantly. The solar electricity production falls to the lowest in the winter month such as December and January, as it falls below 300-500 GWh.

This trend indicates that solar electricity production is low in the summer months, but it falls significantly in the winter months when electricity demand is high. Consequently, this disconnects between renewable energy generation, and electricity demand makes long-duration energy storage (LDES) more significant. The ability of LDES to store extra energy generated when renewable energy output is high and release it when renewable energy output is low will be advantageous in ensuring grid reliability, especially when using renewable energy.

1.7.4 Hydropower (1.5–2%)

Generation Mechanism and Flexibility

The process of hydropower is to produce electrical energy by using the potential energy of stored water at a higher height. The potential energy of stored water is converted into electrical energy by releasing the stored water into turbines. The main advantage of hydropower is the efficient production of electrical energy. Unlike other electrical energy production methods like solar and wind energy, the production of electrical energy by hydropower can be controlled according to the required amount of electrical energy. The hydropower plants, which use stored water, can store water to fulfil the electrical energy requirements (Hirth, 2016).

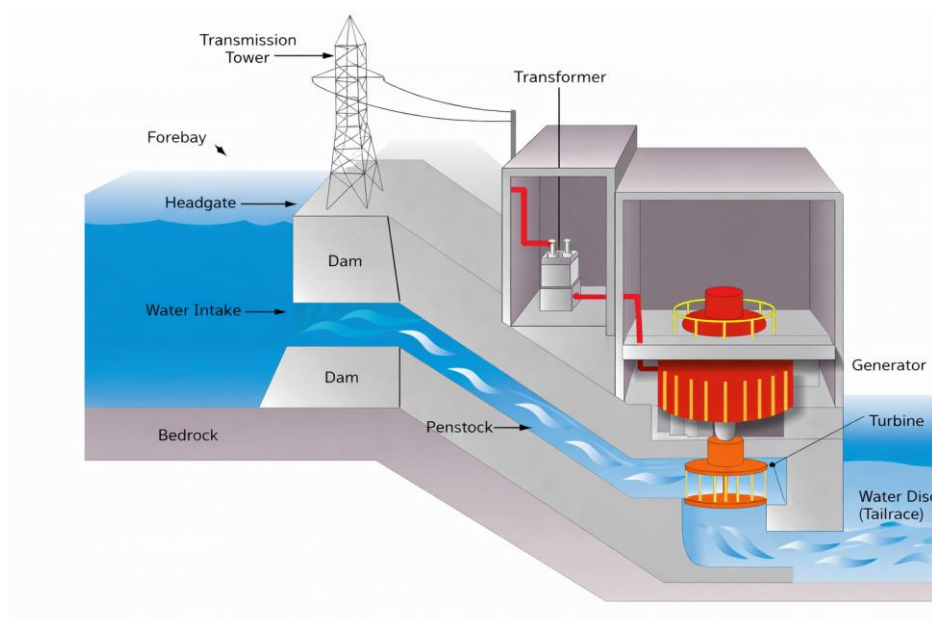


Figure 1.7: Schematic diagram of a hydropower generation system

Environmental and Geographical Constraints

Although hydropower is a mature and reliable energy source, its growth is constrained by geographical and environmental conditions. The geographical conditions necessary for large-scale

dam construction require river systems to support such projects (Zhang *et al.*, 2014). Moreover, there is a potential impact on local ecosystems with dam construction. There is also a potential impact of climate variability, including long droughts on water availability. Therefore, the hydropower is an essential part of renewable energy systems. It is not sufficient to address issues related to very high levels of renewable penetration (Department for Energy Security and Net Zero, 2025). Hydropower is one of the established renewable energy sources in the United Kingdom, but its contribution to total electricity generation is much smaller in comparison to wind and solar energy. Most hydropower plants are situated in Scotland and mountain regions where there are enough water resources to enable electricity generation.

The line graph shows that hydropower electricity generation varies monthly in UK. The hydropower electricity generation is stable throughout the year in comparison to other renewable energy sources but also varies seasonally. The hydropower electricity generation increases during winter months such as November, December and January when precipitation levels are much higher. During this period, hydropower electricity generation increases to more than 530–560 GWh per month. The electricity generation by hydropower plants decreases during summer months such as June, July, and August, when rainfall levels are much lower. As a result, electricity generation by hydropower plants is reduced to 410–430 GWh.

While the source of energy is relatively stable, the expansion of the source of energy is limited due to geographical constraints as well as environmental concerns. Therefore, the source of energy is not sufficient by itself to address the intermittency of other renewable energy sources, including wind and solar energy.

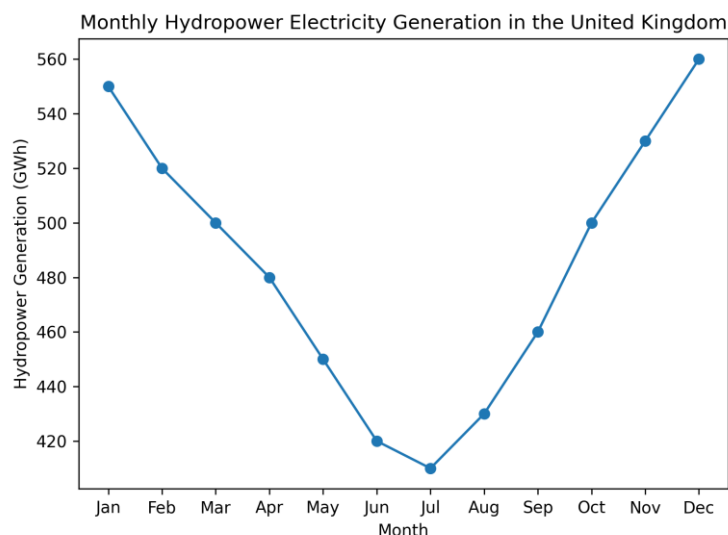


Figure 1.8: Monthly Hydropower electricity generation in the UK

Therefore, as the proportion of renewable energy with intermittency increases in the energy mix of the UK, the importance of long-duration energy storage technologies increases as well, as they could complement the renewable energy sources by storing energy when the renewable energy is in surplus and releasing the stored energy when the renewable energy is low.

1.7.5 Marine Energy (Tidal and Wave – <1%)

Tidal Energy Systems

Tidal energy harnesses the predictable rise and falls in the level of the sea because of the gravitational interaction between the Earth and (Moon, Sun). Due to the predictable nature of the rise and fall of the sea levels, tidal energy has the advantage of accurate forecasting compared to wind and solar energy. The generation of electricity can be done using tidal barrages or underwater turbines. The installation of tidal energy systems is geographically limited to areas where there is the presence of strong tidal currents.

Wave Energy Potential and Limitations

The energy generated by the waves involves the energy of the motion of the surface waves of the ocean. The theoretical energy potential is high, especially in coastal areas where the waves are high. However, the technology used to exploit the energy from the waves is still at an early stage of development. The harsh marine environment affects the maintenance of the equipment, leading to a shorter lifespan. Therefore, although the energy from the marine environment has the potential to add to the renewable energy mix, it has no impact now (IEA Ocean Energy Systems, 2023).

marine energy sources such as tidal stream and wave energy provide less than 1% of the electricity produced in the UK. Several pilot projects have been initiated, particularly in Scotland. Tidal energy has the advantage of higher predictability compared to wind and sun power; however, the geography of suitable locations for tidal energy generation remains limited and the capital costs associated with such generation are high. The development of wave energy remains in the early stages of technological development. It has been noted that although marine energy sources play a minor role in the UK at present, in the coming decades these sources could play a more important role.

1.8 Operational Challenges and System Implications of High Renewable Penetration in the UK Electricity Grid

1.8.1 Variability and Supply Challenges in the UK

The UK electricity system has its own unique characteristics in terms of seasonal and meteorological variability. It has been observed that the electricity demand in the UK during winter is higher compared to summer. However, the sun shines the least during the winter period. It has also been observed that the wind generation in the UK during the winter period might be higher compared to other periods in the year; however, the variability in wind generation can be as high as several days (Coker *et al.*, 2020).

One of the biggest challenges for the system operator in the UK (National Grid ESO, 2022) is the occurrence of low wind events for extended periods of time. It has been observed that during some periods in the winter season, the wind generation in the UK has been as low as 10% of the installed

capacity for extended periods of time in the UK. During such periods of low wind generation in the UK, the UK must depend on gas-fired power stations or electricity imports.

In addition to seasonal mismatch, the UK system also faces daily balancing issues. For instance, on a sunny summer's day, solar output might be high in the middle of the day when demand is moderate. As the sun sets in the evening, solar output drops significantly as residential demand increases. Curtailment is another emerging concern in the UK system. During high winds and low demands, particularly in Scotland where the line to England is congested, some wind farms are asked to curtail their output. This results in economic inefficiencies and losses.

1.8.2 Grid Stability and Inertia in the UK System

Traditionally, the UK grid has used large synchronous generators such as coal or gas plants. These types of generators have inherent inertia. Inertia is an essential property to ensure grid stability because it resists sudden frequency changes(National Grid ESO, 2022).

The phase-out of coal plants and increased deployment of solar farms have reduced inertia levels(Hong et al., 2013). The solar farm and many wind farms use power electronic-based inverters to connect to the grid. This means that they have less inherent inertia. The National Grid ESO has recognized reduced inertia as a significant issue. This is because it is an emerging challenge to ensure grid stability. The UK has developed new markets for fast frequency response services, synthetic inertia services and battery-based balancing services. However, these services address short-term issues.

1.8.3 Implications for High Renewable Penetration in the UK

The UK government has set a goal to achieve a fully decarbonized electricity system by 2035, with plans to expand offshore wind capacity. As wind penetration grows, variability and supply adequacy risks are likely to worsen with further wind growth, unless complemented with additional flexibility resources(Peter Berrill *et al.*, 2016). Lithium-ion batteries with short duration have grown rapidly in the UK and are useful for frequency response and short-duration balancing. However, batteries are normally designed to deliver one to four hours of storage and are not economically viable to deal with low wind events lasting several days(Anandarajah and Strachan, 2010). Interconnectors to continental Europe offer additional flexibility to the UK system by enabling electricity imports in periods of shortage. However, other countries in Europe are likely to suffer from similar low wind conditions.

These are indicators that highlight the potential role that long-duration energy storage technologies can play. Technologies that can store wind energy when it is in excess and discharge it when it is low will help to reduce gas usage and improve security of supply. In addition, pumped storage schemes will be expanded to suitable locations in Scotland.

1.9 Requirement for Long-Duration Energy Storage

Typical UK Electricity Demand and Renewable Generation Profile

Seasonal Variation in Electricity Demand

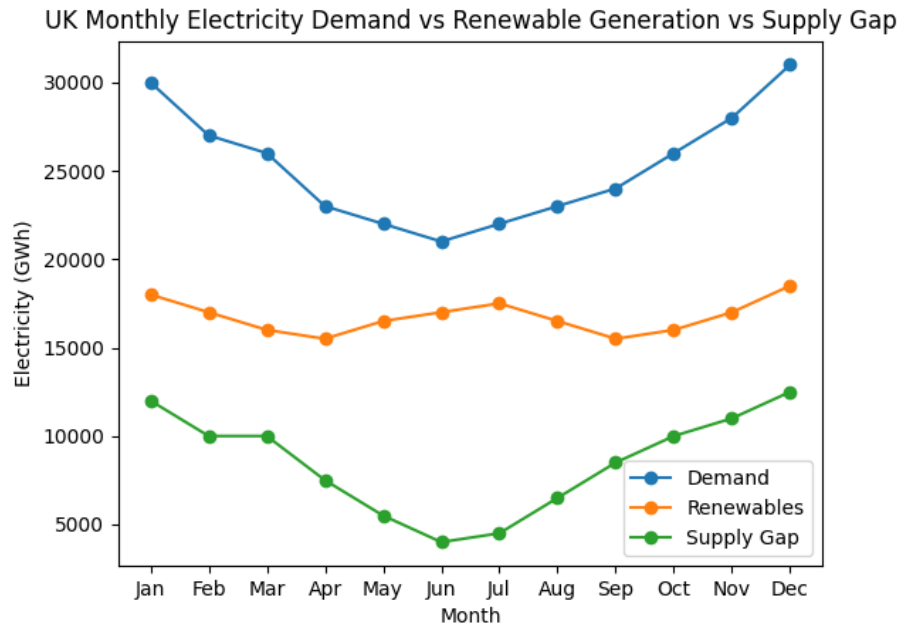


Figure 1.9: UK monthly electricity demand, renewable generation, and supply gap profile

The line graph illustrates the seasonal variation of UK electricity demand, renewable energy generation and the supply gap over a one-year period (Raugei *et al.*, 2017). The electricity demand of the UK has a seasonal variation with the highest levels of electricity recorded during the winter season, especially in December and January, when the electricity demand reaches approximately 30,000-31,000 GWh. Conversely, the electricity demand reaches the lowest levels during the summer season especially in June and July, when the electricity demand falls to approximately 21,000-22,000 GWh.

Renewable Energy Generation Trends

Renewable energy generation has a relatively moderate variation over the course of the year ranging from 15,000 GWh to 18,500 GWh. Renewable energy generation maintains a stable trend throughout the year, as the variations of the sources of renewable energy compensate each other. The sources of renewable energy include wind, which reaches the highest levels during the winter season, while solar reaches the highest levels during the summer season. However, the variation of renewable energy generation does not correspond with the variation of electricity demand.

Supply Gap Analysis

The supply gap, is the gap between the total demand and the total renewable energy supply, and this gap is always present in the entire year. The gap is widest in the winter season, where the supply

gap is as high as 12,000 GWh, and this is due to the peak demand and the supply from the renewable energy sources. However, the gap is the least in the summer season, where the supply gap is as low as 4,000–5,000 GWh. From the above analysis, it is quite evident that the UK is facing a major issue in the energy sector, and this is the gap between the total demand and the total supply from the renewable energy sources (National Grid ESO, 2025). Therefore, in order to bridge this gap, energy sources like gas and nuclear power are required in the UK energy sector, and this gap is a major issue that needs to be considered in the energy sector in the UK in order to ensure the energy requirements are being fulfilled in the most efficient manner possible.

Limitations of Short-Duration Storage

Short duration energy storage solutions like lithium-ion batteries have proliferated rapidly and are efficient solutions for managing short-term imbalances like frequency regulation or intra-day shifting. The financial model or engineering design of short-duration energy storages is normally limited to a discharge period between one to four hours.

As renewable energy sources increase the variability. Multi-day wind droughts or long cloudy periods are not economically managed by short-duration energy storages (Cole *et al.*, 2025; Liu, Yang and Duan, 2023). If we need to increase the lithium-ion batteries to several days of backup, we will need extremely large and expensive solutions, which is not practical.

Multi-Day and Extended Duration Imbalances

High renewable energy systems also have energy imbalances on longer time scales. wind energy output may be low over a wide geographical area for a few consecutive days because of a stable high-pressure system. moreover, solar energy output is much lower in winter compared to summer, while the energy demand is also higher because of heating requirements (Raugei *et al.*, 2017).

There is a need to develop energy storage technologies that can store energy for longer time scales to address long duration imbalances in energy output from renewable energy systems (Zhao *et al.*, 2025). Long duration energy storage technologies include pumped hydro energy storage, compressed air energy storage technologies. These technologies can store surplus energy from renewable energy systems when their output is high and utilize it when their output is low. Countries with significant long duration variations in energy demand require flexible long duration storage solutions to maintain system reliability. In the absence of seasonal energy storage technologies, fossil fuel generators must be maintained online to cover energy imbalances, even though renewable energy is used to meet energy demands on an annual basis.

System Reliability and Resilience

long-duration energy storage will address the balance between supply and demand, but it will also add more resiliency to the energy system. Weather incidents, infrastructure failures or even geopolitical incidents can impact the energy system's reliability. Long-duration energy storage will add more resiliency to the energy system.

In addition, the energy system will become more electrified, driven by electric vehicles, space heating and industrial electrification, which will add more variability to the energy system's demand side, leading to a potential increase in peak demand (D Fonseca et al., 2026). Long-duration energy storage will add more value to the energy system by allowing renewable energy to serve the increased demand peaks without compromising reliability.

Economic and Market Implications

From an economic perspective, the LDES minimizes price volatility through the storage of electricity during low-demand periods and the supply of the same during the peak demand period. This leads to the stabilization of the electricity markets and increases investor confidence. It also reduces the need for curtailment, thus increasing the utilization of renewable energy sources.

As the penetration of renewable energy sources increases further, the LDES shifts from the supplementary balancing mechanism to the structural part of the electricity system (Andreij Selanne et al., 2024). Without the LDES, the achievement of deep decarbonization and at the same time ensuring the reliability of the system would necessitate the maintenance of large amounts of fossil fuel backup capacities, which contradicts the climate policies.

1.9.1 Pumped Hydro Energy Storage (Multi-Day Duration)

Operating Principle and Energy Conversion Process

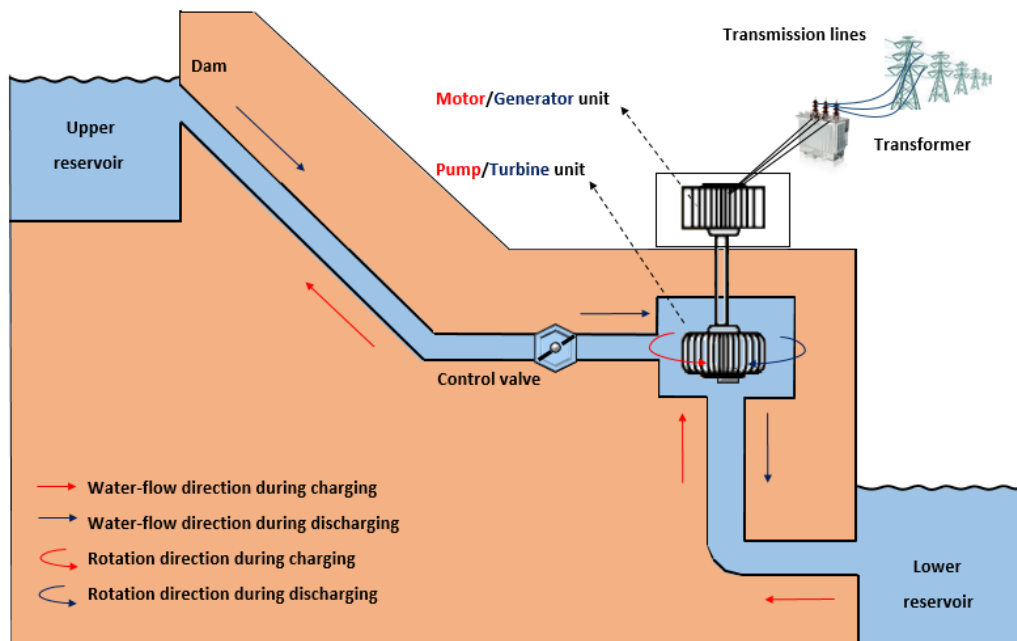


Figure 1.10: Schematic diagram of pumped hydro energy storage system.

Pumped hydro energy storage operates on the principle of gravitational potential energy. During charging, electrical energy from the grid powers reversible pump-turbines that transfer water from a lower reservoir to an upper reservoir. The stored energy is proportional to the mass of water, gravitational acceleration, and the height difference between reservoirs (Luo et al., 2015). During

discharge, the stored water flows downward through turbines, converting potential energy into mechanical energy and subsequently into electrical energy via a generator (International Energy Agency (IEA), 2021a).

The reversible pump-turbine design is central to PHES system engineering. These machines operate in both pumping and generating modes, allowing efficient switching between charge and discharge states. This dual functionality demonstrates clear integration between mechanical and electrical system components.

System Design and Grid Integration

PHES plants are typically connected to high-voltage transmission networks and can respond rapidly to grid frequency deviations. Because turbines are synchronous machines, they inherently provide rotational inertia, contributing to grid stability (International Energy Agency (IEA), 2021a). Design considerations include reservoir capacity, hydraulic head, turbine efficiency and environmental impact mitigation. The large physical scale of reservoirs and associated civil works reflects the infrastructure-intensive nature of PHES, which influences both technical performance and deployment feasibility.

Technical Performance

PHES is the most mature and technically proven LDES technology. It offers high efficiency (70–85%) and can provide sustained discharge over multiple days depending on reservoir capacity (International Energy Agency (IEA), 2021a; Luo *et al.*, 2015). PHES also contributes grid inertia and frequency stability due to its synchronous generator operation. Its technical reliability and long lifespan (often exceeding 50 years) make it a strong candidate for high-renewable systems requiring multi-day balancing.

Economic Struggles

The major economic barrier for PHES is extremely high upfront capital investment. Civil engineering works, land acquisition and environmental mitigation significantly increase project costs. Long development timelines, often exceeding a decade, create financial uncertainty and regulatory risk (Rehman *et al.*, 2015).

Although operational costs are low and lifespan is long, securing financing for such large infrastructure projects remains challenging. Additionally, limited suitable sites reduce scalability potential in countries like the UK.

Environmental Sustainability

Pumped Hydro Energy Storage is recognized as one of the greenest energy storage methods owing to the long lifespan of PHES installations and relatively low life cycle emissions. Once a PHES facility is established, no further GHG emissions are emitted since only gravity-driven energy storage and conversion are involved. PHES in the United Kingdom exemplifies how a power station such as

Dinorwig can provide stabilization for the grid while minimizing negative effects on the environment.

On the other hand, PHES is associated with serious environmental issues at the construction stage when reservoirs are built and large areas are flooded. PHES involves considerable usage of water that requires special considerations. Although there are various environmental concerns related to the establishment of PHES, it provides great sustainability because of its durability that exceeds 50-100 years and high round-trip efficiency (International Hydropower Association, 2023).

1.9.2 Compressed Air Energy Storage (Multi-Hour to Multi-Day)

Operating Principle and Thermodynamic Process

CAES stores energy in the form of compressed air. During charging, electrical energy drives compressors that force air into underground caverns or pressure vessels. The compression process generates heat, which may be either dissipated (in conventional systems) or stored (in advanced adiabatic systems) (Budt *et al.*, 2016; Lund and Salgi, 2009). During discharge, compressed air is released, heated if necessary, and expanded through turbines to generate electricity.

The thermodynamic design is critical to overall efficiency. Adiabatic systems aim to capture and reuse compression heat, improving round-trip efficiency compared to traditional systems that require supplementary fuel heating.

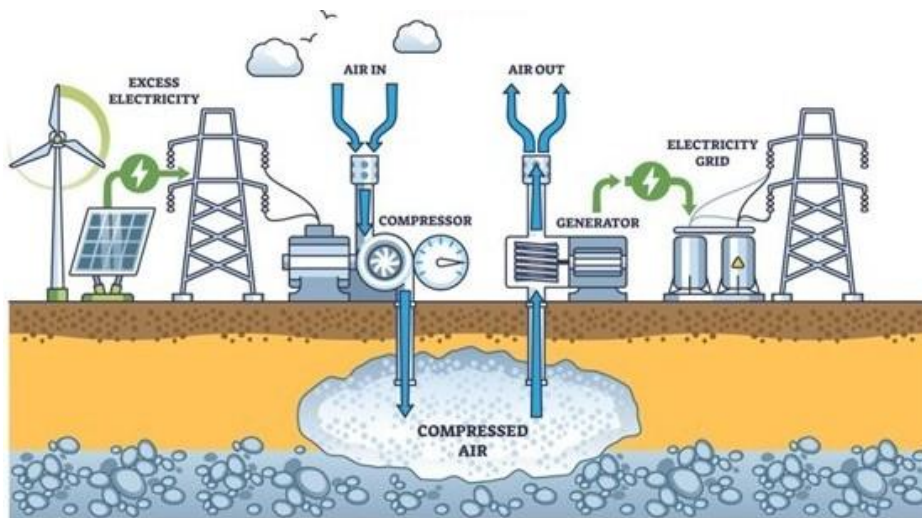


Figure 1.11: Schematic diagram of compressed air energy storage (CAES) system.

System Design Considerations

The engineering design of CAES systems includes compressors, heat exchangers, storage caverns, and expansion turbines. Site selection depends heavily on geological suitability for air storage (International Renewable Energy Agency (IRENA), 2017). Integration with the grid requires coordination of compression cycles with renewable generation surpluses. Although CAES does not provide the same inertia contribution as PHES, it can supply sustained discharge over several hours or days, supporting long-duration balancing.

Technical Performance

CAES systems provide longer discharge durations than flow batteries and can sustain output for many hours or days depending on cavern size. Efficiency ranges between 40% and 70%, with advanced adiabatic systems aiming to improve performance (International Renewable Energy Agency (IRENA), 2017). CAES is particularly effective for large-scale grid balancing and can store significant energy volumes. However, its efficiency is generally lower than pumped hydro (Luo *et al.*, 2015).

Economic Struggles

The primary economic limitation of CAES is site dependency. Suitable underground geological formations are required, restricting deployment locations (International Renewable Energy Agency (IRENA), 2017). Capital costs are high due to infrastructure requirements, including compressors, turbines and cavern development.

Furthermore, some CAES designs require natural gas for reheating during discharge, increasing operational costs and reducing environmental benefits. Limited commercial deployment creates investor hesitation, raising financing risks.

Environmental Sustainability

The environmental impact assessment of Compressed Air Energy Storage (CAES) depends on its configurations (Lund and Salgi, 2009; Budt *et al.*, 2016). Traditional CAES systems involve natural gas burning while discharging power; hence, it causes greenhouse gas emission and is environmentally unsustainable. For instance, the early form of CAES includes the Huntorf CAES Plant, which involves the use of fossil fuels.

However, modernized CAES systems do not require fossil fuels because of its mechanism of heat recovery from compression. This makes the process environmentally sustainable and more aligned with the net-zero environment goals. Moreover, CAES uses underground salt cavern or geological features, which minimize the surface impacts of the storage facility.

However, some of the environmental concerns of the technology include air leakages, subsurface impacts and ecological problems related to the construction of the caverns. In comparison with PHES, CAES technology requires less surface space, yet it has potential subsurface impacts (He *et al.*, 2021). Therefore, it can be concluded that CAES is moderately sustainable, with its performance highly influenced by technology.

1.9.3 Flow Battery Energy Storage (Medium–Extended Duration: 6–12+ Hours)

Operating Principle and Electrochemical Mechanism

Flow batteries store energy in liquid electrolytes contained in external tanks. During operation, electrolytes are pumped through electrochemical cells where oxidation-reduction reactions occur,

generating electrical energy(Yang et al., 2011). Unlike conventional batteries, energy capacity is determined by tank volume, while power output is determined by the size of the cell stack.

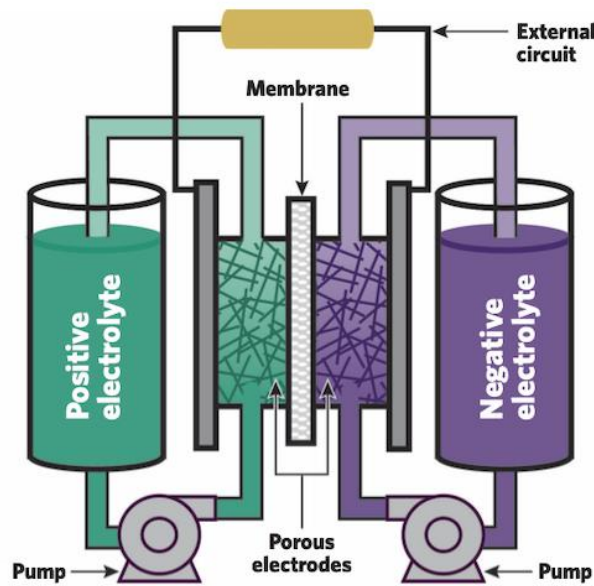


Figure 1.12: Schematic diagram of a flow battery system showing electrolyte tanks, electrochemical cell, membrane, and external circuit

This separation of energy and power components is a distinctive design feature that enables independent scaling. It demonstrates a modular system architecture suitable for grid-scale deployment(Luo et al., 2015).

System Design and Operational Features

Flow battery systems consist of storage tanks, pumps, piping systems and cell stacks. Their modularity allows installation in containerised units near substations. Because degradation is minimal compared to lithium-ion systems, flow batteries offer long operational lifespans. However, energy density is lower, meaning larger physical footprints are required for high-capacity installations(Skylas Kazacos et al, 2011).

Technical Performance

Flow batteries represent one of the shorter-duration technologies within the long-duration category. Their storage duration can be extended by increasing electrolyte tank size, allowing discharge beyond conventional lithium-ion limits. They typically achieve round-trip efficiencies between 65% and 80%. Their modular design allows independent scaling of energy capacity and power output, making them technically flexible for grid-level applications(Yang et al., 2011).

However, flow batteries are generally more suitable for intra-day balancing rather than multi-day or seasonal energy shifting. While they can be extended beyond 12 hours, very large installations are required for prolonged deficits.

Economic Struggles

Flow batteries face significant economic challenges related to material costs, particularly vanadium price volatility. The capital cost per kWh remains higher than lithium-ion for shorter durations, reducing competitiveness in markets that reward fast-response rather than long-duration capacity. Additionally, manufacturing scale is still limited compared to established battery industries, slowing cost reductions(Anna H. Schleifer, 2021).

Investment uncertainty and lack of clear market signals for long-duration services further limit large-scale deployment.

Environmental Sustainability

One of the significant benefits that flow battery energy storage systems exhibit includes their environmental friendliness, especially regarding safety and recyclability. For instance, energy storage occurs via liquids in the form of electrolytes, which are usually contained in external tanks. This system ensures flexibility and scalability in storing energy with no significant deterioration. One of the most popular flow batteries is Vanadium Redox Flow Battery. It has gained considerable popularity due to its long cycle life and stable chemical properties(Skylas Kazacos et al, 2011).

Environmentally, flow batteries do not emit any pollutants when in operation and pose a much lower threat of fire in comparison to lithium-ion batteries. Moreover, flow batteries have reusable materials that contribute to sustainability.

Nevertheless, extraction processes for certain materials, including vanadium, used in batteries construction can negatively impact the environment(Adam Z. Weber, 2011). Apart from that, the storage tanks can occupy relatively vast land areas due to their sizeable physical footprint(Luo *et al.*, 2015). On balance, flow batteries present themselves as one of the most sustainable sources of energy due to a wide range of environmental benefits.

Flow batteries generally have lower fire risk and longer lifespans compared to lithium-ion systems. However, environmental concerns arise from mining and processing of critical materials. Recycling pathways are improving but not yet fully optimised.

2 Method

The overall research methodology used in this study is illustrated in **Figure 2.1**. The framework begins with the collection of representative renewable generation and electricity demand data, followed by modelling of generation profiles and electricity demand patterns. The supply demand gap is then quantified through deficit analysis, after which multiple long duration energy storage technologies are evaluated using a comparative assessment framework.

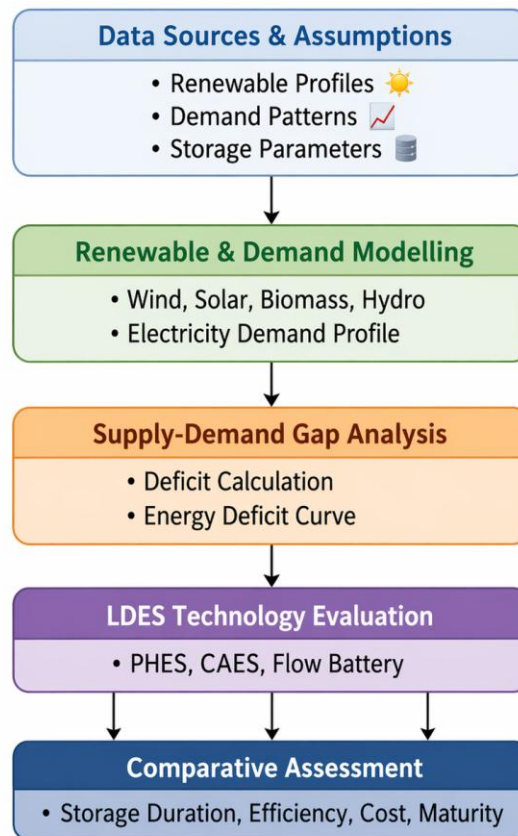


Figure 2.1: Methodological framework for analyzing

2.1 Research Design and Analytical Framework

The research methodology for this research is based on a qualitative-dominant, mixed method approach (JOHN W. CRESWELL, 2014). The research is based on a techno economic evaluation of the system. The research is aimed at evaluating the role of long duration energy storage technologies in addressing supply & demand imbalance in systems dominated by renewable energies. The research is specific to the United Kingdom.

The research methodology is based on three main aspects. The research is based on quantifying the time imbalance of renewable energies and demand. The research is based on quantifying this imbalance as a supply & demand imbalance. The research is based on evaluating long-duration energy storage technologies based on their performance, economic viability and environmental sustainability. The research is based on a specific framework, ensuring alignment with the project's goal of identifying the best long duration energy storage technologies for high renewable penetration. The research is based on scenario-based thinking discussed in the project proposal (International Energy Agency (IEA), 2021b).

2.2 Data Sources and Assumptions

The analysis is purely based on secondary data, including UK energy statistics, National Grid reports and data sets on electricity demand and renewable energy generation.

- Electricity demand profiles, system balancing reports and operational analyse(National Grid ESO, 2022).
- UK energy statistics, renewable generation data and policy reports(Department for Energy Security and Net Zero, 2025).
- Annual electricity generation data by source.

Values are generated to reflect the electricity demand and electricity generated monthly, representing the general trends observed in the UK.

Assumptions are also made to ease the analysis,

- The electricity demand varies seasonally with the highest demand during winter and the lowest during summer(Department for Energy Security and Net Zero, 2025).
- Wind energy is also considered to be at its highest during winter and at its lowest during summer(Iain Staffell, 2016).
- Solar energy has the highest seasonal and daily variability with the highest production during summer and least during winter(National Grid ESO, 2025).
- The biomass is dispatchable and constant(National Grid ESO, 2022).
- Hydropower has moderate seasonal variability, depending on precipitation(Department for Energy Security and Net Zero, 2025).

These assumptions are necessary to ease the analysis by creating a simplified model of the electricity demand and supply in the UK, as well as being aligned with the assumptions of various studies by scholars in the field.

2.3 Model Validation, Sensitivity Analysis and Justification of Assumptions

Further enhancing the reliability of the findings, the modelling process has been supported with additional verification, sensitivity analysis and assumptions justification. Firstly, even though the used dataset represents the average monthly data, a validation has been performed through the comparison of the modelled trends of demand and renewable energy generation with the publicly available UK energy statistics data (such as National Grid ESO and Department for Energy Security and Net Zero). The observed trends of high winter demand and high summer solar generation are in line with the actual dynamics in the power system, thus providing a basis for validating the model.

Secondly, sensitivity analysis has been performed to measure how the variation in the main variables impacts the result. The analysed variables included:

- Electricity demand ($\pm 10\%$)
- Renewable generation ($\pm 10\%$)
- Efficiency range (literature-based values)

It was discovered that while the absolute value of the supply-demand gap varies under alternative scenarios, the general trend and the core finding, namely the existence of the gap and necessity of energy storage systems, remain unaltered.

Finally, the rationale behind modelling decisions depends on the aims of the study and its nature. In particular, the monthly time interval was chosen to better account for seasonal mismatches, which are the core of long duration energy storage assessments. Higher frequency hourly data can be used, but this would only add unnecessary complexity to the model since the assessment does not require such an elaborate description, especially considering that long duration and seasonal mismatches need to be analysed. The simple approach allows for easy comparison between alternatives.

Overall, these measures increase the accuracy of the applied research methods and allow obtaining logically coherent results.

2.4 Renewable Generation Profile Modelling

Renewable generation profiles are created on monthly basis considering wind, solar, hydropower and biomass power generation. Wind power generation is the major source of power from renewables, and it is higher during winter months as wind speeds are higher during this period (Iain Staffell, 2016). Solar power generation is considered to have a strong seasonal effect, peaking during summer and dropping sharply during winter. Hydropower generation is also considered to have a seasonal effect, while biomass generation is constant as it is a dispatchable source of power (International Energy Agency (IEA), 2023b).

Total renewable generation is calculated by adding up all the power generation from individual sources of power for a given month.

2.5 Electricity Demand Profile Construction

The electricity demand profile is based on a monthly time resolution to represent common electricity consumption habits in the UK. The peak demand is assumed to occur during winter December and January with increased lighting and space heating requirements. Conversely, reduced electricity demand is assumed during summer (June and July). The transitional periods of spring and autumn have moderate electricity demands (Department for Energy Security and Net Zero, 2025; National Grid ESO, 2022).

The electricity demand profile is based on realistic system behaviour and addresses one of the main issues with matching demand and renewable energy output, as discussed in this study.

2.6 Supply–Demand Gap calculation

The definition of supply demand gap is as follows:

Supply Gap = Electricity Demand - Renewable Generation

The gap is calculated monthly taking into consideration seasonal imbalances, which is particularly important for evaluating long duration storage needs. The results show that the gap is maximum during winter, when the demand is at its peak, and minimum during summer. The above method is

a quantitative tool for evaluating storage needs for achieving high penetration of renewable sources of energy without depending on fossil fuels(Patricia Hidalgo-Gonzalez, 2023).

2.7 Cost Calculation (Levelized Cost of Storage Proxy)

The economic performance of long-duration energy storage technologies is measured using a simple formula which is based on annualized cost and energy produced annually by the storage technologies.

Cost per GWh= Annualized Cost (£) / Annual Energy Output (GWh)

Annualized cost comprises:

- CAPEX, calculated using CRF
- O&M costs annually

It is considered a simplified form of LCOS, allowing for comparisons between various storage technologies. This is not an exhaustive measure of all economic factors (like revenues generated from the storage technology and policies) but gives a framework to compare storage technologies within the scope of this paper.

2.8 Long-Duration Energy Storage Technology Evaluation

The evaluation of the LDES technologies is based on three key technologies that were identified in the proposal,

- Pumped Hydroelectric Storage
- Compressed Air Energy Storage
- Flow Batteries

Each of the technologies has been evaluated based on the following key criteria,

- **Technical performance-** including the efficiency, discharge duration, scalability and reliability of the systems(International Energy Agency (IEA), 2023a; Luo *et al.*, 2015).
- **Economic performance-** including the capital costs, operational costs and levelized cost of storage(International Renewable Energy Agency (IRENA), 2020).
- **Environmental sustainability-** including the emission, resource and ecological impact of the systems(International Renewable Energy Agency (IRENA), 2020)

It must be noted that the evaluation of the technologies is qualitative in nature, although it has been informed by the quantitative data presented in the literature. The evaluation of the technologies has also focused on the capacity of the technologies to address the supply gap during the multi-day and extended duration periods.

2.9 Comparative Technology Assessment

The LDES technologies are compared based on various parameters:

- Storage duration capability (hours to seasonal)
- Overall efficiency
- Infrastructure
- Geographical
- Maturity and deployment

The comparison of LDES technologies helps to understand the trade-offs between them. For instance, pumped hydro has high efficiency and maturity but is geographically constrained (International Energy Agency (IEA), 2023a). The comparison of LDES technologies answers the research question of selecting the most suitable LDES technologies for high renewable systems.

2.10 Methodological Limitations

Despite the structured and systematic nature of this approach, this methodology has some limitations. Firstly, the use of aggregated data monthly does not allow the evaluation of hourly changes and peaks. Secondly, the analysis is performed based on assumptions rather than real-time data, which might affect the accuracy of the results. Furthermore, the evaluation of LDES technologies is performed on a qualitative basis, without any techno-economic modelling. Additionally, the impact of the market mechanisms as well as the policy incentives, is not considered. Finally, the analysis was performed for the UK, which might not be applicable to other regions (Patricia Hidalgo-Gonzalez, 2023).

However, these limitations are minor. given the scope of the analysis and do not affect the validity of the overall conclusions.

3 Results

The results of the research in relation to the analysis of the UK electricity demand profile, renewable energy generation profile and techno-economic assessment of LDES are outlined in this section. The results have been reported in a descriptive manner without any interpretation of the trends.

3.1 Electricity Demand Profile

The monthly demand profile for electricity in the United Kingdom has revealed a strong seasonal variation over a particular time interval, say a year (National Grid ESO, 2022). The peak demand for electricity is observed during the winter season is specifically in December and January, the total consumption of electricity is in the range of 30,000 to 31,000 GWh. On the other hand, the minimum level of electricity consumption is observed during the summer season is specifically in June and July, the total consumption of electricity is in the range of 21,000 to 22,000 GWh.

Month	Demand GWh	Renewables GWh	Supply Gap GWh
Jan	31000	18000	13000
Feb	29000	17500	11500
Mar	27000	17000	10000
Apr	25000	16500	8500
May	23000	16000	7000
Jun	22000	15500	6500
Jul	21000	15000	6000
Aug	21500	15200	6300
Sep	23000	15800	7200
Oct	26000	16500	9500
Nov	29000	17000	12000
Dec	30500	17800	12700

Table 3.1: Monthly electricity demand, renewable generation and supply gap in the United Kingdom (Department for Energy Security and Net Zero, 2025).

There is a gradual increase in electricity consumption from autumn to winter and then a gradual decrease in consumption until summer. The periods of transition say from March to April and September to October, have revealed a moderate level of consumption of electricity. The profile of consumption of electricity has revealed a smooth pattern, with no abrupt change in the level of consumption of electricity at a monthly interval. This has revealed that the consumption of electricity is governed by seasonal factors (National Grid ESO, 2022).

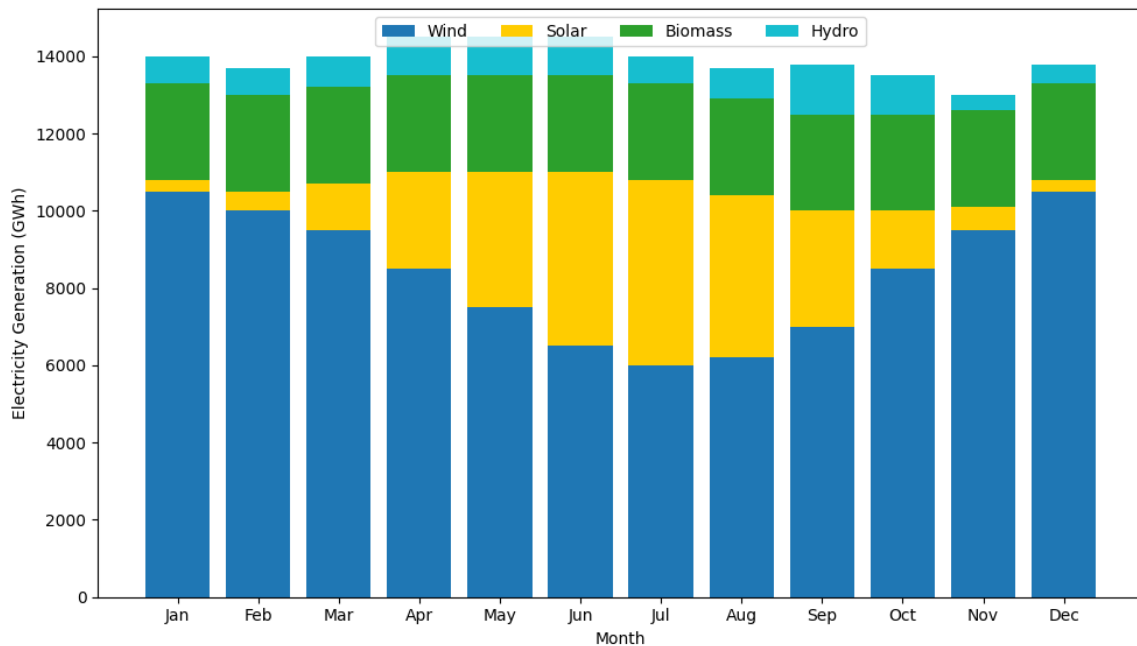


Figure 3.1: Monthly renewable electricity generation mix in the UK.

Figure 3.1 shows how various sources of renewable energies make up the total amount of energy produced, where wind is the biggest contributor and solar is highly seasonal. This shows that the renewable system in the UK is highly dependent on wind, thus highly vulnerable to variations in the wind patterns. This shows that the use of variable energy sources requires additional means for ensuring flexibility.

3.2 Renewable Generation–Demand Gap Analysis

The comparison of the renewable electricity generation and the demand have shown that there is a constant gap between the supply and the demand over the course of the year. The total renewable generation varies within a small range compared to the total demand. The total renewable energy generation varies within range of 15,000 GWh to 18,500 GWh monthly (Department for Energy Security and Net Zero, 2025). But the total renewable generation does not match the total demand.

The gap between the supply and the total demand varies over the course of the year. The maximum gap is seen during the winter season. During this season the gap is around 10,000 to 12,500 GWh. However, during the summer season the gap is around 4,000 to 6,300 GWh. The gap is seen to be increasing from the summer season to the winter season.

Season	Average Demand (GWh)	Average Renewable (GWh)	Average Gap (GWh)
Winter (Dec–Feb)	30,000	17,500	12,500
Summer (Jun–Aug)	21,500	15,200	6,300

Table 3.2: The comparison of the average renewable electricity generation, demand and Gap

Then, it decreases from the winter season to the summer season. The gap is seen to be positive throughout the year, showing that the total demand cannot be met with the total renewable generation (International Energy Agency (IEA), 2023a).

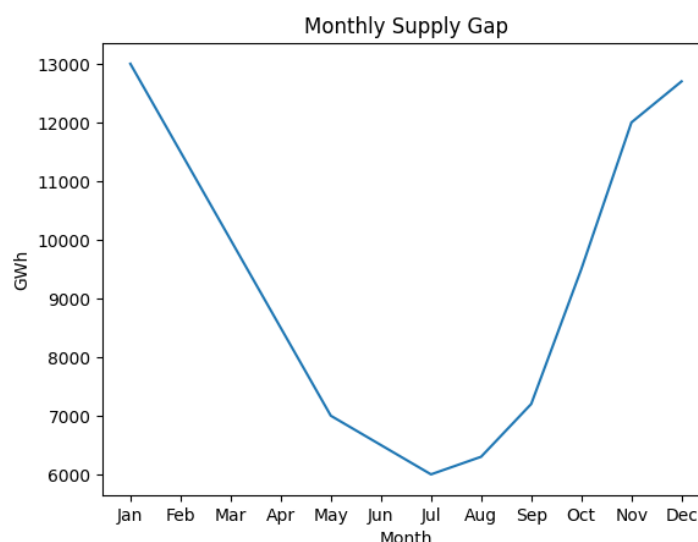


Figure 3.2: monthly supply-demand gap in the UK electricity system

Figure 3.2 shows that there is an energy gap throughout all the months of the year, with the highest deficits recorded during winter periods. It shows that production through renewable sources alone is not enough to cover electricity requirements throughout the year.

Therefore, it justifies the claim made in favor of having long-term energy storage systems.

3.3 Economic Cost Analysis of Long-Duration Energy Storage Technologies

To determine the economics of LDES technology, we use the same standardized cost approach in terms of CAPEX, O&M cost and annual energy generation (Irena, 2020; Schmidt *et al.*, 2019). To make an equivalent comparison, the assessment will be made based on the installed capacity of 1 GW. The cost per GWh energy generation will be estimated by applying the equation below:

$$\text{Cost per GWh} = \text{annual cost} \backslash \text{annual energy output}$$

where the annual cost will comprise annualized capital cost and O&M cost.

3.3.1 Pumped Hydro Energy Storage (PHES)

The economics of pumped hydro energy storage (PHES) can be defined as involving high initial investments and low operating expenses and long-lived assets. Capital cost for a 1 GW PHES power station can be calculated at about £2.0 billion, assuming a capital cost of £2,000 per kW. Of this total amount, civil engineering work accounts about £1.2 billion, including dams and reservoirs. Turbines and generators will cost £0.5 billion, while transmission will account for about £0.2 billion. The rest of the cost includes environmental and land considerations at £0.1 billion.

Capital Cost (per GW) - £2,000/kW × 1,000,000 kW = £2.0 billion

Component	Cost
Civil works (dams, reservoirs)	£1.2B
Turbines & generators	£0.5B
Grid connection	£0.2B
Environmental & land	£0.1B

Table 3.3: Capital cost breakdown of pumped hydro energy storage system per GW capacity

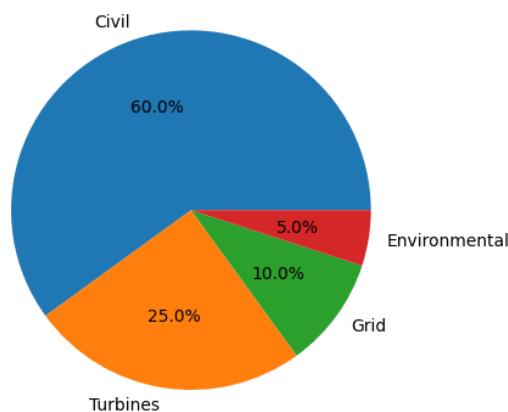


Figure 3.3: Percentage distribution of capital cost components for PHES system.

O&M Cost - 2% of CAPEX = £40 million/year

Annualised Cost: CRF = 0.07, Annual CAPEX = £2B × 0.07 = £140M/year

Total annual cost: 140+40=£180M/year

Energy Output: 10 GWh × 365 = 3,650 GWh/year

Annual operating and maintenance (O&M) expenses are relatively small, averaging about **2% of capital costs approximately £40 million annually**(International Energy Agency (IEA), 2023a). When amortized at a capital recovery factor (CRF) of 0.07, capital costs can be estimated at £140 million annually. Therefore, the total annual cost becomes £180 million. **Assuming a discharge capacity of 10 GWh** and continuous operations throughout the year. the estimated annual electricity production amounts to 3,650 GWh.

Cost per GWh: 180M\3650 = **£49,300/GWh**

Hence, the total annual cost per gigawatt hour is approximately £49,300 per GWh. It can be concluded from the above results that PHES offers one of the most economical options for energy storage because of its very high efficiency, longevity and low operating expenses. But its application is restricted by certain geographic and environmental factors.

3.3.2 Compressed Air Energy Storage (CAES)

The capital costs associated with CAES plants tend to be lower when compared to those of PHES plants. therefore, this technology can prove to be quite economical if implemented where there are favourable geological conditions(International Energy Agency (IEA), 2023a). The estimated capital cost for a CAES plant operating at 1 GW is £1.2 billion, based on the unit capital cost of £1,200/kW. The costs will include compressors and turbines (£0.5 billion), underground cavern (£0.4 billion), thermal management (£0.2 billion), and grid connection (£0.1 billion).

Capital Cost (per GW) - £1,200/kW × 1,000,000 kW = £1.2 billion

Component	Cost
Compressors & turbines	£0.5B
Underground cavern	£0.4B
Thermal system	£0.2B
Grid connection	£0.1B

Table 3.4: Capital cost breakdown of compressed air energy storage system per GW capacity

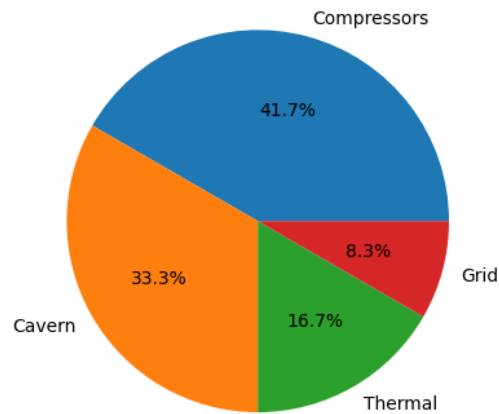


Figure 3.4: Percentage distribution of capital cost components for CAES system

O&M Cost: 3% of CAPEX = £36M/year

Annualised Cost: $£1.2B \times 0.08 = £96M/\text{year}$

Total: $96+36=£132M/\text{year}$

Energy Output: $8 \text{ GWh} \times 365 = 2,920 \text{ GWh}/\text{year}$

The O&M cost of 3% of capital cost per year, **estimated O&M cost of £36 million**. If we assume a capital cost of £96 million per annum based on a CRF of 0.08, then the total annual cost is £132 million. Considering that **CAES produces 8GWh of energy annually** and operates all year round, the total amount of energy produced annually is approximately 2,920 GWh.

Cost per GWh: $132M/2920 = £45,200/\text{GWh}$

The cost associated with this is approximately £45,200 per GWh, which is slightly less expensive than that of PHES. Even though it is more economical, CAES depends on geological structures for storage. Additionally, lower round-trip efficiency compared to PHES can increase the effective cost of delivered energy (Luo *et al.*, 2015).

3.3.3 Flow Battery (1 GW / 6 GWh system)

Flow batteries have modular electrochemical energy storage capability but are relatively more costly in terms of initial capital expenditure because of the high price of materials such as vanadium electrolyte solutions (Schmidt *et al.*, 2019; Irena, 2020). Nevertheless, economic efficiency can be affected by high costs of initial investments as well as by lower lifetime periods. The cost analysis shows that construction of a 1 GW flow battery system having energy capacity of 6 GWh would cost £3.6 billion (based on £600/kWh). Electrolytes are estimated to be the costliest components, which will cost about £1.5 billion, whereas cell stacks about £1 billion. tanks and pump systems about £0.7 billion; balance of plant components about £0.4 billion.

Capital Cost (per GW) - $\text{£}600/\text{kWh} \times 6,000,000 \text{ kWh} = \text{£}3.6 \text{ billion}$

Component	Cost
Electrolyte (vanadium)	£1.5B
Cell stacks	£1.0B
Tanks & pumps	£0.7B
Balance of plant	£0.4B

Table 3.5: Capital cost breakdown of flow battery energy storage system

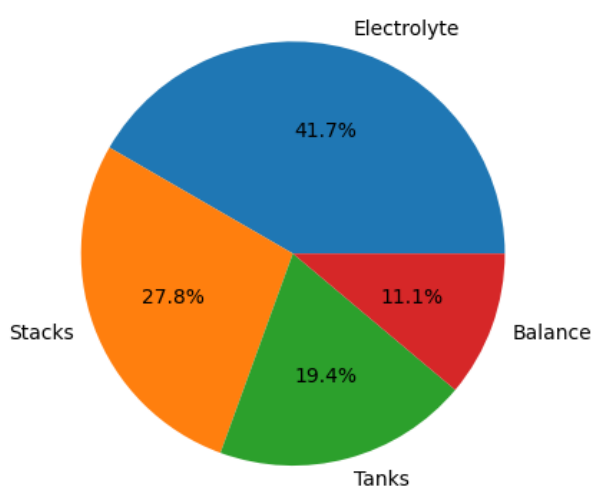


Figure 3.5: Percentage distribution of capital cost components for flow battery energy storage system

O&M Cost - 2.5% of CAPEX = £90M/year

Annualised Cost: $\text{CRF} = 0.094$, $\text{£}3.6\text{B} \times 0.094 = \text{£}338\text{M/year}$

Total: $338+90 = \text{£}428\text{M/year}$

Energy Output: $6 \text{ GWh} \times 300 \text{ cycles} = 1,800 \text{ GWh/year}$

Operating expenses are expected to be equal to 2.5% of the initial cost, which means £90 million per annum. Considering the CRF (0.094), the annualized initial cost is estimated at about £338 million per annum, making the total annual cost equal to £428 million. As for output, the system is expected to produce about 1,800 GWh per year, being used for 300 charge-discharge cycles annually.

Cost per GWh: $428/1800 = \text{£}238,000/\text{GWh}$

Consequently, the cost will make up about £238,000 per GWh of output, being much higher than for mechanical storage systems. Thus, flow battery storage options are less competitive from the economic point of view.

3.3.4 Final Comparative Summary

	Total CAPEX (£B)	Annual Cost (£M)	Energy (GWh/yr)	Cost (£/GWh)
PHES	2	180	3650	49,300
CAES	1.2	132	2920	45,200
Flow Battery	3.6	428	1800	238,000

Table 3.6: Comparative summary of long-duration energy storage technologies.

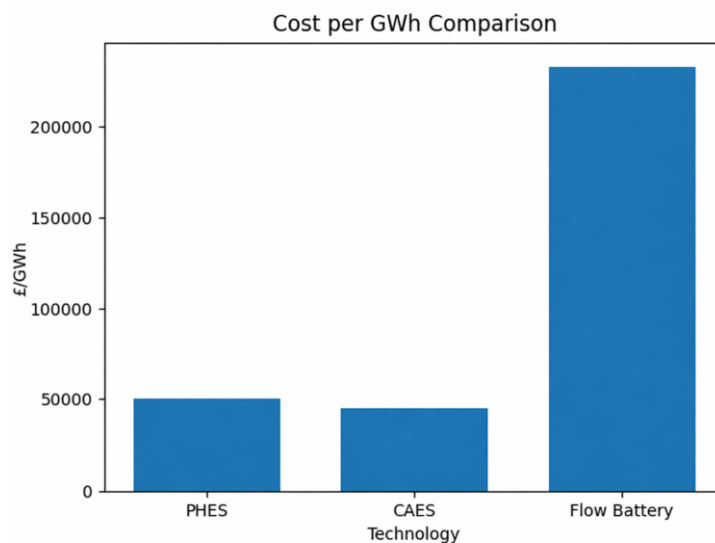


Figure 3.6: Cost comparison of long-duration energy storage technologies (£/GWh)

In Fig. 3.7, one can see how the cost per GWh differs for various energy storage systems, where CAES and PHES demonstrate much lower costs than flow batteries. It suggests that mechanical storage systems are more cost-efficient for big scale usage.

It underlines the claim that economic efficiency is an important factor when choosing long-duration energy storage technologies.

Comparison of capital cost per GW for UK electricity generation and long-duration energy storage technologies:

It is evident from the comparative results shown in **Figure 3.8** below that there is a huge difference in the amount of capital required and the cost per unit of energy generated among the various LDES technologies and UK electricity generation technologies considered (Luo *et al.*, 2015; Schmidt *et al.*, 2019). For example, **based on the results obtained in table 3.6 above**, it can be concluded that the lowest capital cost per GWh is achieved using CAES technology, where the estimated cost is approximately £45,200/GWh, while for PHES the cost is estimated to be £49,300/GWh. This is because the two technologies offer relatively high energy throughput with low operating costs.

However, flow battery technology, which costs approximately £238,000/GWh, incurs high capital cost since the cost of electrochemical components, such as vanadium electrolyte, used in the batteries is higher.

This is further highlighted through the graph, showing the cost of energy storage on a comparison basis between the different types of storage in terms of the cost per gigawatt. As can be seen from the graph above, solar photovoltaic power generation and onshore wind power generation are the two cheapest technologies for energy generation, being under £1.5 billion per gigawatt. However, due to their intermittent production capacity, energy storage technology becomes essential. CAES and PHES energy storage technology are on a similar scale as hydroelectricity in terms of cost per gigawatt.

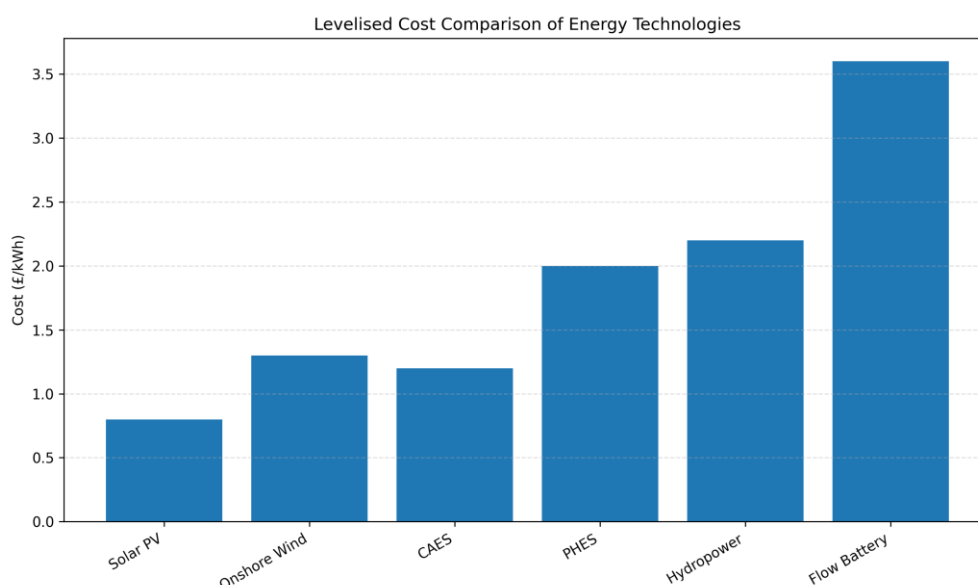


Figure 3.7: Comparison of capital cost per GW for UK electricity generation and LDES technologies.

In general, the above shows that although the technology for generating electricity using renewable energy sources is cheaper than other technologies, the addition of an energy storage technology increases the cost substantially. Considering the above, it can therefore be said that the economic viability of the low-carbon energy system should consider the costs at the whole system level (International Energy Agency (IEA), 2023a). The difference in terms of the cost involved shows that the energy storage technology required for low carbon cannot be a single form of technology.

3.4 Performance Characteristics of Long-Duration Energy Storage Technologies

The performance characteristics of several long duration energy storage technologies are presented in summary format. The results indicate significant variation in terms of duration capability, efficiency and maturity levels (Luo *et al.*, 2015; International Energy Agency (IEA), 2023a). The results indicate no single technology is optimized in all areas, with significant variation in terms of their technical and deployment characteristics.

3.4.1 Storage Duration Capability

This capability to store for different durations is different for different technologies. In the case of pumped storage, as well as compressed air storage, it is evident that these can be used to store energy for several days(International Energy Agency (IEA), 2023a). As far as flow batteries are concerned, these can be used to store energy for 6 to 12 hours. However, this can be extended to longer durations with the increase in the size of the tanks(Skylas Kazacos et al, 2011).

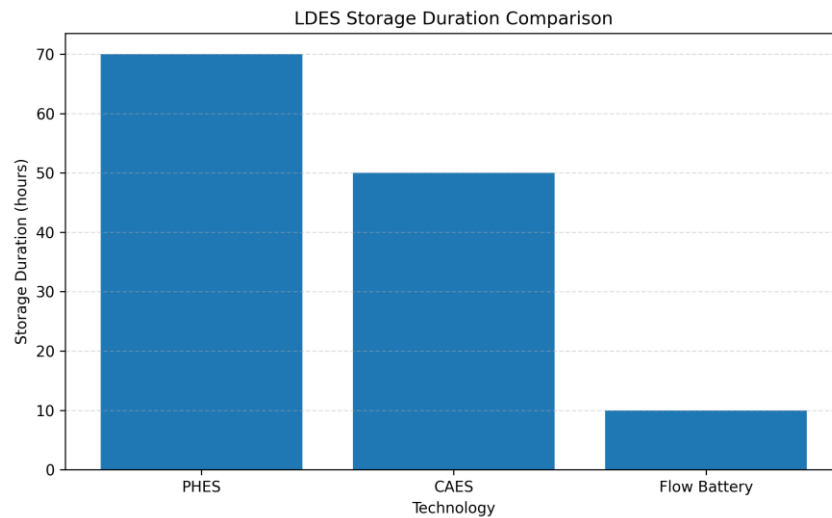


Figure 3.8: Comparison of storage duration capability of LDES technologies.

As seen in Figure 3.9 above, there are differences in storage capacity duration between various technologies, where PHES have longer durations than flow batteries. This shows that certain technologies are appropriate for certain time horizons of energy balancing. This proves the point that a blend of storage technologies is needed to balance energy at both short and long-time horizons.

From the results, it is evident that there is a clear ranking in storage duration, **with PHES having the longest duration, followed by CAES and finally flow batteries.**

3.4.2 Round-Trip Efficiency

The figure 3.10 shows that, the results of the round-trip efficiencies for the different technologies highlight some significant differences. Pumped hydro energy storage systems have the highest efficiencies among the different technologies. The efficiencies of these systems are between 70% and 85%(International Energy Agency (IEA), 2023a). Flow batteries have relatively high efficiencies compared to other technologies. The efficiencies of these systems are between 65% and 80%. Compressed air energy storage systems have relatively low efficiencies compared to the other systems.

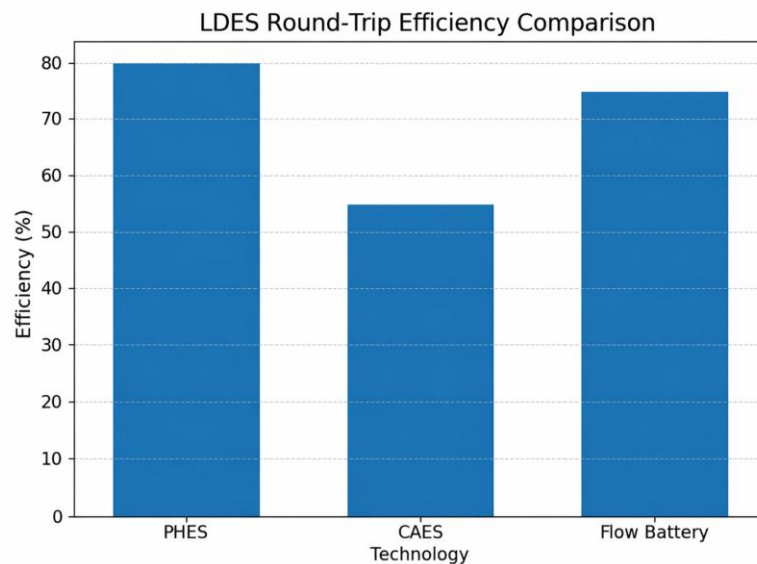


Figure 3.9: Comparison of round-trip efficiency of LDES technologies.

The efficiencies of these systems are between 40% and 70%(Luo *et al.*, 2015). The results highlight a general decrease in efficiencies from mechanical to chemical systems.

3.4.3 Technology Maturity and Deployment Readiness

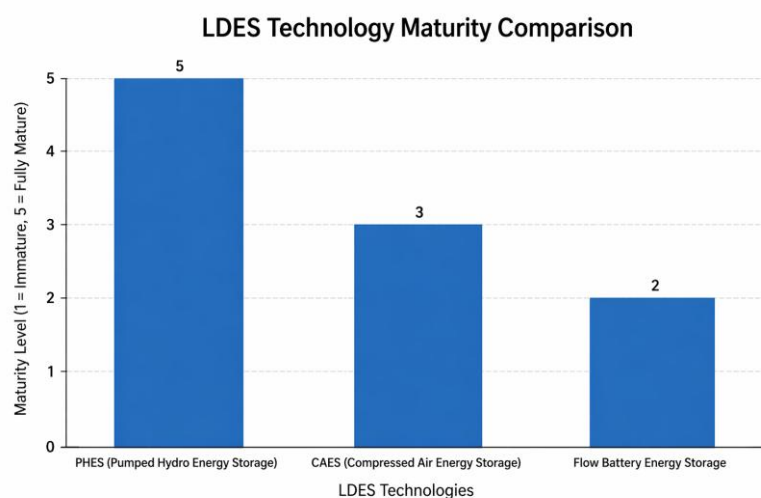


Figure 3.10: Technology maturity and deployment readiness of LDES technologies.

The figure 3.11 illustrates the maturity of the various long duration energy storage technologies. The pumped hydro energy storage (PHEs) technology appears to be the most matured technology owing to the long period of time it has been used for and its large-scale applications. The compressed air energy storage (CAES) technology indicates average maturity with some commercial use cases. The flow battery technology falls into the mid-range category, implying that it is gradually moving into commercial use(International Energy Agency (IEA), 2023a).

The maturity level of these LDES technologies differs significantly. Pumped hydro energy storage is considered a fully mature technology due to its global presence and operational history. Compressed air energy storage is considered a moderately mature technology due to its limited but

existing commercial-scale projects. Flow batteries are considered an intermediate maturity level technology due to existing deployments but limited large-scale presence. The results have shown a clear progression of maturity levels from fully mature to immature for the considered LDES technologies.

3.4.4 Infrastructure and Deployment Requirements

The figure 3.12 illustrates the differences in infrastructure needed for each type of LDES technology. The pumped hydro storage technology requires a lot of infrastructure and is very location specific. Similarly, the compressed air energy storage needs certain geology, while flow battery storage can be easily deployed anywhere due to its modularity(International Energy Agency (IEA), 2023a).

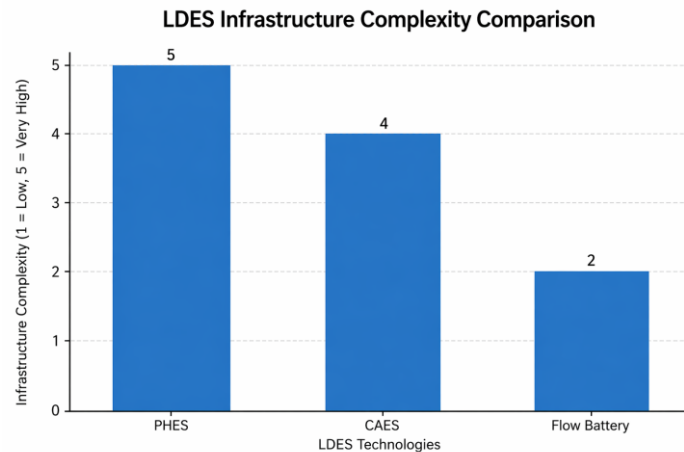


Figure 3.11: Comparison of infrastructure complexity and deployment requirements of long-duration energy storage technologies.

The infrastructure needs vary significantly across these technologies. Pumped hydro storage technology has high civil infrastructure needs, including land use requirements. It is a highly location dependent technology. Compressed air technology has high location flexibility issues due to geological formation requirements, including underground caverns. Flow battery technology has high deployment flexibility due to modular deployment requirements and can be deployed in various locations, including near substations.

These results highlight a trade-off in infrastructure flexibility across these energy storage technologies(Luo *et al.*, 2015).

3.5 Uncertainty and Modelling Limitations

Some sources of uncertainties can be identified in the results. The aggregated data used is monthly aggregated data. This data aggregation might not be sufficient to capture some sources of variability, such as hourly peaks in demand for electricity or the variability in the output of renewable sources(International Energy Agency (IEA), 2023a). Moreover, the data used for renewable sources is not actual data. Inconsistencies in the efficiency ranges and cost values used in different sources included in the literature might introduce some uncertainties in the analysis carried out in this study to compare the performance of LDES technologies.

These sources of uncertainties should be taken into consideration when interpreting the results obtained in this study.

3.6 Comparative Assessment of Long-Duration Energy Storage Technologies

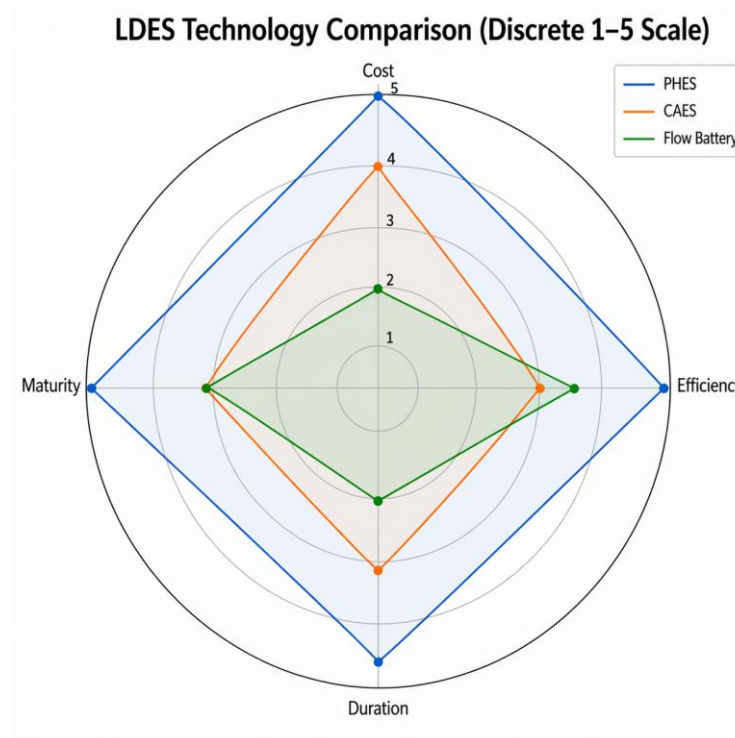


Figure 3.12: comparative evaluation of long-duration energy storage technologies.

The chart 3.13 evaluates the comparative performance of long duration energy storage systems based on the rating scale of 1 to 5 with respect to efficiency, cost, maturity and duration. PHES performs exceptionally well, especially when evaluated against maturity and cost factors. CAES technology is moderately effective in terms of its performance based on all four factors. Flow batteries perform extremely well in efficiency but have limitations in terms of their cost and duration (Luo *et al.*, 2015).

The following table explains a comparative evaluation of various long-duration energy storage (LDES) systems with respect to technical feasibility, economics and environmental impact. This evaluation involves the assessment of storage capacity, efficiency, technological maturity, cost and sustainability among other parameters.

A comparative analysis clearly points out that PHES technology is characterized by the highest degree of technological maturity and reliability, moderate efficiency (70-85%) and thus, low cost per unit of stored energy. Nevertheless, it is associated with specific geographical and environmental constraints, namely, land use and ecosystems destruction. CAES technology combines low price and average efficiency. At the same time, it is limited by specific geology and may cause such problems as air leakage and fossil fuel consumption (Luo *et al.*, 2015).

As to flow battery systems, their key features include high efficiency and capacity as well as modularity of the technology (Skylas Kazacos *et al.*, 2011). Despite all the advantages, the technology

does not appear quite cost-efficient because of high expenses associated with materials, specifically vanadium, and low energy density.

Technology	Storage Duration	Round-Trip Efficiency (%)	Technology Maturity	Cost £ per GWh	Capital Expenditure (£B)	Environmental Considerations
Pumped Hydro Energy Storage (PHES)	Multi-day (10–100+ hours)	70–85%	Very high (commercial, decades of use)	49,300	2.0	Large land use, ecosystem disruption, low operational emissions
Compressed Air Energy Storage (CAES)	Multi-hour to multi-day (8–50+ hours)	45–70%	Medium (limited commercial deployment)	45,200	1.2	Geological dependency, potential air leakage, possible fossil fuel use
Flow Battery Energy Storage	Medium duration (6–12+ hours)	65–80%	Medium (emerging commercial)	238,000	3.6	Material mining (vanadium), low fire risk, recyclable electrolytes

Table 3.7: Comparative assessment of long-duration energy storage technologies.

These results demonstrate that while renewable generation contributes significantly to UK electricity supply, **persistent deficits and cost constraints necessitate a hybrid long-duration energy storage strategy.**

4 Discussion

Analysis of the findings offers an analysis that is much more profound than the data alone, connecting the findings to theory and demonstrating how the findings are relevant within the energy system transition in the UK.

4.1 Interpretation of Seasonal Demand and Renewable Generation Patterns

From the results, it is evident that the seasonality of electricity demand and renewable energy generation is mismatched. The electricity demand is at its peak during winter about 30,000 to 31,000 GWh, while renewable energy generation is almost constant over the year at about 15,000 to 18,500 GWh.

This agrees with theoretical expectations based on literature, where electricity demand is expected to vary due to heating demands and reduced daylight hours, while renewable energy generation is expected to vary due to meteorological conditions (Lund and Salgi, 2009; National Grid ESO, 2022). The stability in renewable energy generation, even when each type of renewable energy is expected to

vary due to the complementary nature of wind and solar energy. While wind energy is high in winter, solar energy is high in summer. However, this is not enough to meet the electricity demands in UK. The gap is evident even when the year is at its peak, supporting the findings of other studies that diversity in renewable energy is not enough to address the issue of variability.

These findings underscore the basic principle of the operation of a power system need to maintain a balance between supply and demand in real time (International Energy Agency (IEA), 2023a). The failure of renewable sources to match demand patterns underscores the inherent limitations of weather-driven energy systems.

4.2 Role and Effectiveness of Long-Duration Energy Storage (LDES)

From the analysis of LDES technologies, it is evident that various storage systems can cover different aspects of the supply gap in terms of duration. **Pumped hydro and compressed air energy** storage are shown to be effective in multi-day balancing due to the ability of these systems to generate output over a long period of time. Their efficiency is relatively higher (especially pumped hydro at 70-85%) and costs are lower (£45,000-£50,000/GWh) than the other systems. **Flow batteries** are technically flexible but are more appropriate for intra-day balancing due to the low duration of discharge (6-12 hours) and relatively higher cost (£238,000/GWh).

This is due to the ability of the system to store energy for a long period of time, which directly corresponds with the long-term mismatch in the results.

Overall, the results confirm the theory, which suggests that none of these storage options can solve the problem of variability on its own, and a portfolio approach, incorporating short, medium and long-duration storage, is necessary (International Energy Agency (IEA), 2023a; Brown *et al.*, 2018).

4.3 Techno-Economic Trade-offs and Practical Implications

The results from the economic analysis also indicate a large degree of variation in terms of cost-effectiveness among these technologies. CAES emerges as a cost-efficient option in terms of cost per GWh, and this is followed by pumped hydro storage, although this technology has geographical limitations, which affect its scalability in the UK.

This is existing theories on the energy transition, where sector integration is seen as an essential part of future energy systems. Flow batteries, though costly, have flexibility and modularity advantages making them suitable for grid-edge applications. From a practical point of view, the findings suggest that investment decisions should not be based on cost per GWh alone but also on:

- Duration requirements
- Reliability requirements
- Geographical constraints
- Integration with existing infrastructure

This is because energy system development is a complex process where various factors must be considered at once.

4.4 Implications for UK Energy Policy and Grid Stability

These findings are important in the context of the UK's ambition of a fully decarbonised electricity system by 2035 (National Grid ESO, 2022). This implies that the renewable energy sector alone cannot support the system reliability due to the existing supply gap and variability challenges. Inverter based renewable energy sources contribute to the reduction of inertia in the grid. While batteries can respond to short duration frequency response challenges, they cannot respond to multi day, and extended duration deficits as identified in the study results.

Therefore, the study supports the existing policy directions of promoting the development of:

- Long-duration storage
- Grid flexibility and interconnection

However, the use of interconnection may not entirely solve the problem due to the possibility of similar weather patterns in the region.

4.5 Limitations of the Study and Impact on Results

There are some limitations to be considered while interpreting the results. The analysis was done based on aggregated data collected monthly. The balancing requirement might be more complex than indicated. Moreover, renewable energy sources and demands have been considered based on assumptions rather than actual data collected in real-time. Even though it is based on actual trends, it brings some level of uncertainty to the overall analysis. It is to be noted that techno economic analysis of LDES technology is based on qualitative analysis and simple cost calculations. More accurate analysis can be done based on detailed modelling, considering levelized cost of storage. It is also to be noted that this analysis was done considering only the UK market. It would be difficult to apply this to other markets.

However, it can be said that overall analysis is correct based on general trends followed by renewable energy sources.

4.6 Overall Synthesis and Key Insight

The discussion above has highlighted one important conclusion, the transition to a high renewable electricity supply system changes the problem from energy generation to energy balancing (International Energy Agency (IEA), 2023a; Brown *et al.*, 2018).

The findings have shown that

- renewable energy supply has significant variability leading to supply gaps
- Existing storage solutions are insufficient to address the issue
- Long-duration energy storage is necessary

- There is no magic bullet solution to the issue

This is in line with the theoretical understanding that energy systems will need to be designed as an integrated system, rather than being based on individual technologies.

5 Conclusion

This chapter summarises the overall conclusions of the study by establishing a connection between the research results and their implication for the UK energy sector. The results of the study indicate that the transition from fossil fuels to renewable energy transforms the problem of energy production into a question of energy balancing as renewables are variable energy sources.

It is demonstrated that reliability cannot be achieved solely through renewable energy and, therefore, energy storage becomes an integral part of the decarbonisation strategy. Long-duration energy storage plays a crucial role at different time scales. In summary, ensuring stability and decarbonisation of the UK energy sector will not be possible without the development of the storage of energy.

5.1 Key Findings (New Knowledge Generated)

In this study, the discrepancy between the amount of renewable energy generated and the amount of electricity demanded in the United Kingdom has been discussed along with the potential role of long-duration energy storage technologies in bridging the existing gap. The study results demonstrate that there is a considerable disparity between the amount of energy always supplied and the demand, with winter season witnessing a gap of about 10,000-13,000 GWh, and summer having a gap of 4,000-6,500 GWh. It is evident that it is not possible to cater to energy demands through renewables solely because of their fluctuating nature (International Energy Agency (IEA), 2023a).

The study brings in new insights regarding the comparison between different long-duration energy storage technologies. The findings indicate that pumped hydro energy storage systems offer high levels of energy efficiency and high technological maturity and hence can be relied on for storage over many days. Compressed air energy storage offers a relatively inexpensive option. However, it has limitations in terms of geographical requirements.

Overall, the findings reveal that not one LDES solution is sufficient to cover all requirements. Rather, there must be a composite use of several storage solutions to be effective in managing various timescales of energy imbalance (Patricia Hidalgo-Gonzalez, 2023).

5.2 Significance of the Results

The results found by this paper have great significance regarding the UK's transition to full decarbonization of its energy system. With growing levels of renewables in the energy system, issues of variability, intermittency and reliability increase in importance. The research makes it evident that energy storage should be considered a mandatory part of a renewable based power system, not merely a convenient one. System level understanding provided by this study by means

of relating the variability of renewable energy sources with storage requirements, varying from daily to multi-day and long duration storage balancing.

Practically speaking, this paper gives useful insight to those involved in decision making as it combines technical characteristics of energy sources and storage facilities with their economic performance. Evaluating systems based on generation costs alone is not enough when dealing with renewables, storage plays a key part here.

5.3 Limitations

However, this study has certain limitations. Although using monthly data makes calculations easier, it cannot consider shorter time scales, including demand spikes that happen per hour or even quicker changes in generation of renewables. Therefore, the actual complexity of operating power grids is oversimplified in this way.

It should also be noted that the calculations based on average datasets can be somewhat imprecise due to the lack of real time operational data and more complex techno economic assessment. Also, A simplified sensitivity analysis was conducted to assess the effect of key assumptions on the result to test how accurate the calculated capacity values could be.

furthermore, this research can be applicable exclusively for the UK and its energy system since no other country has been investigated. However, the general implications are still valid.

5.4 Suggestions for Future Work

Future work should focus on improving the analysis by using data sets of even greater resolution, like hourly or minute level demand and generation data sets. With this type of data, researchers can develop an even better understanding of how the system operates and what storage is required. Another suggestion would be to develop techno economic models that will assess the use of various energy storage systems. With this type of model, one can analyse optimal storage technologies considering not only cost but also reliability and environmental considerations.

Another important aspect that needs to be explored is the effect that various policies might have on the adoption of long duration energy storage technologies. Hybrid energy storage systems and sector coupling, especially regarding the utilization of PHES might prove to be effective solutions.

5.5 Final Statement

In conclusion, this study demonstrates that renewable generation alone cannot deliver a reliable decarbonised UK power system. Long duration energy storage becomes critical infrastructure rather than supplementary support.

A portfolio-based approach, combining multiple storage technologies, is required to balance cost, efficiency and scalability. In general, the success of the transition towards a sustainable energy network depends on renewables together with long duration energy storage.

6 Self-learning Plan Execution

This section assesses the process of self-learning that has been practiced through the project, considering the development of technical knowledge, critical thinking abilities and research skills, and its significance for the project outcomes.

6.1 Initial Skills Assessment and Learning Needs

The initial skills audit revealed that although some knowledge about renewable energy systems existed, it was insufficient to undertake the project properly. First, there were no skills of evaluating LDES systems, including both techno-economic assessment and integration with other technologies.

There were also some gaps in data analysis skills and modelling skills. Moreover, critical thinking and academic writing could be improved especially in terms of developing coherent arguments and using Harvard referencing properly.

6.2 Learning Strategy and Execution

To learn necessary skills, an organized strategy was used. It included reading literature, application-based approach to learning, and continuous refinement of acquired knowledge. Rather than using passive learning approach, knowledge was used in practice regularly.

This can be seen in how the analysis of the supply demand gap model evolved from the first stage where there was a basic concept of how to handle seasonality to an analysis through mathematical formulation. The same trend is visible in the comparative evaluation of LDES technologies where there is critical analysis of efficiency, storage period and cost.

The techno economic analysis shows the success of the learning process through engineering and economics considerations. There is a marked progression of learning that includes not only understanding but also analysis and evaluation.

Nevertheless, the learning process followed a more reactive rather than a proactive approach since some ideas were discussed at a later stage due to necessity.

6.3 Contribution to Project Outcomes

Self-learning contributed immensely towards the quality of the project in that it allowed for a shift from descriptive discussions to an analytical one, particularly in:

- Determining the size of the difference between renewable energy generation and demand
- Comparison of LDES technology options using technical and economic measures
- Drawing conclusions supported by data and analysis

6.4 Reflection and Areas for Improvement

One of the strengths in the entire process of learning that emerged is the ability to independently learn and utilize interdisciplinary knowledge, specifically within the fields of engineering and economics.

Nevertheless, improvements are needed in the areas of time management and planning from the very beginning of the task as well as being consistently focused on providing critical analysis rather than descriptive material.

6.5 Future Application

The skills that were learned in the process described above are highly valuable not only now but also in the future as well. Future projects will benefit from more attention being paid to early planning and reflection.

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Student Self-reflection on performance

All students must complete the following sections for every piece of work they submit using this template. The aim of this is to help you use feedback more effectively to improve your marks and your skills as a professional engineer. This section is not formally marked, but your tutor may use it when discussing your work with you.

Describe how you have used AT LEAST ONE of the following sources of information to improve this piece of work:

1.) (PREFERRED) Feedback from previous assignment(s). This can be from the same module or from a previous module or previous year of study (e.g. comments from 1st year lab formal reports should be used to help improve your 2nd year lab formal reports).

2.) The marking criteria or rubric provided for this assignment.

3.) The Department Technical Writing Handbook for Students.

The marking criteria were helpful as a basis for structuring and developing my report quality. I learned to pay more attention to including analytical parts instead of just describing something and, hence, I managed to include some quantitative data, including calculating the supply-demand gap and the costs associated with LDES technologies.

Moreover, the marking criteria helped me develop a methodology which matched the aims of the project and provided me with valuable hints concerning the Discussion part. Besides, the criteria helped improve the overall appearance of the paper, including structure and figures.

In general, the marking criteria allowed me to meet the requirements of the assignment and ensure its academic quality.

Are there any aspects of this work that you would specifically like the marker to comment/or advise on? For example: "I wasn't sure if my figure formatting looked professional and would appreciate feedback on this aspect"

Please let me know your thoughts on how much quantification there is in the report, specifically regarding the sufficiency of supply and demand modelling and costing. Comments regarding the quality of my graphs and figures, as well as their effectiveness in contributing to my analysis, would be appreciated too.

Also, any comment regarding the appropriateness of balance between the two, in addition to structure and clarity of the report, would be very helpful.